

Amplitudes_2pi	code/init/lowPhoton/twoPi_production/gamma2pi_Amplitudes.f90		
inMedium_delta_width	logical	.false.	turn on/off the in-medium-width of the delta
inMedium_delta_potential	logical	.false.	turn on/off the in-medium-potential of the delta
inMedium_nucleon_potential	logical	.false.	turn on/off the in-medium-potential of the nucleon
inMedium_pion_potential	logical	.false.	turn on/off the in-medium-potential of the pion
buuPotential	logical	.true.	use buu potentials, else constants

AnalyzeSpectra	code/analysis/analyzeSpectra.f90		
realID	logical, dimension(1:122)	.false.	Switch on/off the output for specific particle IDs from the real particles vector
pertID	logical, dimension(1:122)	.false.	Switch on/off the output for specific particle IDs from the pert particles vector

angular_distribution	code/collisions/phaseSpace/winkelVerteilung.f90		
deltaPWave	logical	.true.	Switch for P-Wave decay of delta in pion nucleon Only relevant for deltas which are produced in pion-nucleon collisions. → see also master_2body Values: • .false.= isotropic in CM-Frame • .true. = $1+3*\cos(\theta)^2$ in CM Frame (θ is angle of producing pion to outgoing pion)
pionNucleon_backward	logical	.true.	Switch for backward peaked pion nucleon cross section: • .true.= use backward peaked distribution • .false.= isotropic
pionNucleon_backward_exponent	real	26.5	Exponent for backward peaked pion nucleon cross section. Distribution= $(\text{coeff} \cdot \cos(\theta))^{\text{exponent}} \cdot (\text{pole} \cdot \sqrt{s}) / \text{pole}$ Only used if pionNucleon_backward=.true. .
pionNucleon_backward_coeff	real	1.9	Exponent for backward peaked pion nucleon cross section. Distribution= $(\text{coeff} \cdot \cos(\theta))^{\text{exponent}} \cdot (\text{pole} \cdot \sqrt{s}) / \text{pole}$ Only used if pionNucleon_backward=.true. .
rho_pipi_nonIsotropic	logical	.true.	Switch for non-isotropic rho → pi pi decay: • .false.= isotropic in CM-Frame • .true. = non-isotropic

NNisotropic	logical	.false.	if .true.: set isotropic nucleon-nucleon elastic cross section
iParam_gammaNVN	integer	3	for $\gamma N \rightarrow V N$ events, this parameter is given to the routine vecmesa and selects there, how $d\sigma/dt$ is calculated. Only if iParam_gammaNVN ≥ 0 the default value of that routine is overwritten. Possible values: • 0: 'old' parametrisation for $\gamma N \rightarrow V N$ (cf. Effenberger PhD): $d\sigma/dt \sim \exp(Bt)$. Slope parameter B according ABBHHM collab, PR 175, 1669 (1968). • 1: Pythia parametrisation: Slope parameter $B=2*b_p+2*b_v+4*s^{**}eps-4.2$ • 2: 'Donnachie, Landshoff' Select t according dsig/dt as given by VecMesWinkel/dsigdt, not by a given slope parameter • 3: as 1, but for rho and $W < \sim 6\text{GeV}$ slope parameter adjusted according CLAS experimental data [Morrow et al, EPJ A39, 5 (2009)] • 4: Muehlich PhD, Appendix E • 5: Rho0 Toy Init • 6: Rho0 Toy Init: Fit to PYTHIA-VMD • 7: Flat (not exp.) cf. VecMesWinkel/vecmesa for a detailed description.
NN_NR_noniso	logical	.false.	If .true., use non-isotropic angular distr. for $NN \rightarrow NR$, according to $d\sigma/dt = b/t^{**}a$.

annihilation code/collisions/twoBodyReactions/annihilation/Annihilation.f90			
model	integer	2	Switch between the models of annihilation: • 1 – string based model, • 2 – statistical model
position_flag	integer	1	Switch between the choices of position of outgoing mesons: • 1 – at the c.m. of the baryon and antibaryon, • 2 – at the antibaryon position

barAntiBar_input code/collisions/twoBodyReactions/baryonBaryon/barAntiBar.f90			
fact_LambdaBar	real	1.	Enhancement factor of $p\bar{p} \rightarrow \Lambda \bar{\Lambda}$ cross section (for larger statistics)
fact_JPsi	real	1.	Enhancement factor of $p\bar{p} \rightarrow J/\Psi$ cross section (for larger statistics)

fact_JPsi_width	real	1.	Enhancement factor of the J/Psi total width (for larger statistics)
useAnni	logical	.true.	Flag whether to perform Baryon-Antibaryon annihilation or not at all

barBar_barBar	code/collisions/twoBodyReactions/baryonBaryon/barBar_barBar.f90		
mat_NR	real, dimension(Delta:F37_1950)	...	Squared matrix elements $M^{*2}/16\pi$ for $N N \rightarrow N R$. See http://arxiv.org/abs/1203.3557 .
mat_DR	real, dimension(Delta:F37_1950)	...	Squared matrix elements $M^{*2}/16\pi$ for $N N \rightarrow \Delta R$. See http://arxiv.org/abs/1203.3557 .
icugnon	integer	1	Switch for nucleon nucleon $\rightarrow$ nucleon nucleon cross sections: • 0=old parametrization • 1=new parametrization (Alexej Larionov, Cugnon)
use_ND_ND_model	logical	.false.	Switch for delta nucleon $\rightarrow$ delta nucleon cross sections: • false=old parametrization • true =one pion exchange model (Effenberger, Buss)
new_NR_NR	logical	.true.	• .false.= Switch off the NR$\rightarrow$ NR improvement (improvement= better NN$\leftrightarrow$NN fit is being used) • only for debugging or comparing
NR_NR_massSHIFT	logical	.false.	• .true.= Shift the srts in NR$\rightarrow$ NR elastic collisions.
oldOset_treatment	logical	.false.	• .true.= Use the old treatment for the Oset Delta width: Put everything into 3-body. • only for debugging or comparing
etafac	real	6.5	Parameter for enhancement of $p n \rightarrow N^*(1535) N$, relative to $p p \rightarrow N^*(1535) N$, in order to enhance eta production in pn collisions. See Calen et al., PRC 58 (1998) 2667.
rhofac	real	1.	Parameter for enhancement of $p n \rightarrow N^*(1520) N$, relative to $p p \rightarrow N^*(1520) N$, in order to enhance rho production in p n collisions.
neufac	real	1.	Parameter for enhancement of $p n \rightarrow N R$, relative to $p p \rightarrow N R$, affecting all resonances.
neufac_roper	real	2.	Parameter for enhancement of $p n \rightarrow N N^*(1440)$, relative to $p p \rightarrow N N^*(1440)$. See http://arxiv.org/abs/1203.3557 .

barBar_barBarMes	code/collisions/twoBodyReactions/baryonBaryon/barBar_barBarMes.f90		
NNpi	logical	.true.	Global switch for the $N N \rightarrow N N \pi$ contibution
NNpi_BG	integer	2	Switch for the $N N \rightarrow N N \pi$ background (s-channel): • 0 = no BG • 1 = BG according to Teis • 2 = BG according to Buss (improves threshold behavior, default) • 3 = BG according to Weil
NNV_BG	logical	.true.	Incude a $N N \rightarrow N N V$ background term, where $V=\omega,\phi$ (in addition to possible resonance contributions).
isofac_omega	real	1.	Isospin enhancement factor for $p n \rightarrow p n \omega$, relative to $p p \rightarrow p p \omega$. Data indicate that this is around 2, while theory predicts even larger values (up to 5). Reference: Barsov et al., EPJ A21 (2004) 521-527.
isofac_phi	real	1.	Isospin enhancement factor for $p n \rightarrow p n \phi$, relative to $p p \rightarrow p p \phi$. Theory predicts values of 3-4, cf.: Kaptari, Kaempfer, Eur.Phys.J. A23 (2005) 291-304.

BarBar_to_barBar_model	code/collisions/twoBodyReactions/baryonBaryon/barbar_to_barbar_model.f90		
couplings_switch	integer	2	Possible values: • 1 = use couplings according to Dmitriev • 2 = use couplings according to Pascalutsa (default)
lambda_cutoff	real	0.6	Cutoff parameter in the form factor for $ND \rightarrow ND$ Possible values: • 0.6 (Dmitriev, default) • 1.2 (Doenges)

baryonPotential				code/potential/baryonPotential.f90
EQS_Type	integer	5	Switch for equation of state for nucleon resonances with spin 1/2. Parameters for nucleon potentials: • 0 = nucleon potential is set to zero • 1 = soft, momentum dependent, lambda = 2.130 (Teis PhD, K = 215 MeV) • 2 = hard, momentum dependent, lambda = 2.126 (Teis PhD, K = 380 MeV) • 3 = soft, momentum independent (Teis PhD, K = 215 MeV) • 4 = hard, momentum independent (Teis PhD, K = 380 MeV) • 5 = medium, momentum dependent, lambda = 2.130 (Teis PhD, K = 290 MeV) • 6 = LDA potential (Birger Steinmueller) • 7 = Deuterium potential Argonne V18 (not usable for eventtypes 'heavyIon' and 'hadron') • 8 = LDA Potential Welke • 9 = Buss PhD, Set#1 (K = 220 MeV, momentum dependent) • 10 = Buss PhD, Set#2 (K = 220 MeV, momentum dependent) • 11 = Buss PhD, Set#3 (K = 220 MeV, momentum dependent) • 12 = Shanghai meeting 2014 (soft, momentum independent, K = 240 MeV) • 13 = slightly modified Cooper potential, central depth = - 67.5 MeV at p=0 (see #14) • 14 = Potential fitted by Cooper et al, Fig. 6 in PRC 47 (1993) 297 • 98 = use pre-stored values • 99 = variable Skyrme : E_bind, p_0, U_0, rho_0 must be defined! NOTES References: • for 1-5, see the PhD thesis of S. Teis, chapter 3.3.2 / table 3.1 • for 9-11, see the PhD thesis of O. Buss, chapter 7.2.3 / table 7.1	
DeltaPot	integer	1	Switch for potential of spin=3/2 resonances: • 0 = no potential • 1 = nucleon (spin=1/2) potential times 2/3 [according to Ericson/Weise book] • 2 = 100 MeV * rho/rhoNull • 3 = nucleon (spin=1/2) potential	

HypPot	integer	1	Switch for potential of hyperons: • 0 = no potential • 1 = nucleon (spin=1/2) potential times (3+S)/3 (i.e. according to the share of the light quarks) • 2 = nucleon (spin=1/2) potential
symmetryPotFlag	integer	0	Switch for the asymmetry term in the nucleon potential. NOTES Possible values: • 0 = none (default) • 1 = linear (strength given by 'dsymm') • 2 = stiffer, $E_{\text{sym}} = E_{\text{sym_rho_0}} * U^{\text{gamma}} = 31. * U^{\text{gamma}}$, gamma=2 • 3 = stiff, linear increasing $E_{\text{sym}} = E_{\text{sym_rho_0}} * U = 31. * U$ • 4 = soft, $U_c=3$, can give negative $E_{\text{sym}} = E_{\text{sym_rho_0}} * U * (U_c - U) / (U_c - 1)$
symmetryPotFlag_Delta	logical	.false.	Switch for the asymmetry term in the Delta potential. NOTES If .true., a symmetry potential will be used also for the Delta (but only if symmetryPotFlag>0). It is closely related to the symmetry potential of the nucleon.
noPerturbativePotential	logical	.false.	Switch for potential of perturbative particles. If .true. then perturbative baryons feel no potential.
rho_0	real	0.16	Nuclear matter density for EQS_Type=99 NOTES • Units : fm⁻³
p_0	real	0.8	momentum for which $U(p_0, \text{rho}=\text{rho}_0)=0$ for EQS_Type=99 NOTES • Units : GeV
U_0	real	0.075	$U(p=0, \text{rho}=\text{rho}_0)$ for EQS_Type=99 NOTES • Units : GeV
bindingEnergy	real	0.016	Nuclear matter binding energy for EQS_Type=99 NOTES • Units : GeV
compressibility	real	0.290	Nuclear matter compressibility for EQS_Type=99 NOTES • Units : GeV

SurfacePotFlag	logical	.false.	Switch for the surface term in the nucleon potential. NOTES • Do not use it together with yukawa!
nLoopReAdjust	integer	10	number of iterations, if density is readjusted (cf. type(nucleus)%ReAdjustForConstBinding) NOTES It is necessary to reiterate (at least for momentum dependent potentials), since we calculate the potential for a given pF and then calculate for the readjusting a new pF.
dsymm	real	0.03	Parameter for symmetry potential in GeV. NOTES Value is only used for symmetryPotFlag = 1

BaryonWidth code/width/baryonWidth.f90			
readTable	logical	.true.	There is a tabulation of the widths saved in buuinput which is used to initialize ('baryonWidthVacuum.dat.bz2'). If you don't want to use this pre-tabulated input, then you can set "readTable=.false.". This is useful for runs on a cluster where you want to minimize input/output. Also it is necessary if the decay channels have been modified (cf. DecayChannels.dat).
writeTable	logical	.false.	This flag determines whether we write out a new tabulation of the widths ('baryonWidthVacuum.dat.bz2'). It will only have an effects if readTable == .false. or reading of the tabulation file fails for some reason.

BaryonWidthMedium_tables code/width/baryonWidthMedium_tables.f90			
minRes	integer	-1000	Read the data table starting at this minimal resonance ID. ONLY FOR TESTING!!!
maxRes	integer	1000	Read the data table up to a maximum resonance ID. ONLY FOR TESTING!!!
inMediumParameterset	integer	2	chooses the parameters for the inMediumWidth: • 1: charges accessible with electron • 2: charges accessible with neutrino (= all)
onlyNucleon	logical	.false.	Only for debugging: only nucleon width is read in.
deltaOset	logical	.false.	Use delta width according to Oset et al. NPA 468 (1987)
extrapolateAbsP	logical	.false.	if true then set absP to maxAbsP if absP is larger

FileNameMask	character(1000)"	The absolute filename of the file containing the in-medium tables. This is only the basename, the final name is given by prej '[FileNameMask].iii.n.s.dat.bz2' where iii is the ID of the particle, n=1,2, and s=1,2 possible values: • if not set, default is '[path_To_Input]/inMediumWidth/InMediumWidth' • if given, but does not contain '/': default is '[path_To_Input]/inMediumWidth/[FileNameMask]' • otherwise: filename is absolute, including path NOTE if you want to use the files 'XXX.iii.n.s.dat.bz2' in the actual directory, give this mask as './XXX'
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BaryonWidthVacuum		code/width/baryonWidthVacuum.f90	
use_cutoff	logical	.true.	• Switch on and off the use of cut off parameters. • These cut-offs are necessary when working with dispersion relations to deduce the real part.
deltaRho_cutoff	real	0.85	• Cut off parameter for the decay of a resonance into delta rho. • Units of GeV
baryon_cutoff	real	2.0	• Cut off parameter for the decay of a resonance into an unstable baryon and a meson. • Units of GeV
meson_cutoff	real	1.6	• Cut off parameter for the decay of a resonance into a baryon and an unstable meson. • Units of GeV
Delta_width	integer	1	Select a parametrization for the Delta width: • 1 = Manley (GiBUU default, cf. Manley/Saleski, Phys. Rev. D 45, 1992) • 2 = Dmitriev (Dmitriev/Sushkov/Gaarde, Nucl. Phys. A 459, 1986) • 3 = Moniz (Koch/Moniz/Ohtsuka, Ann. of Phys. 154, 1984) • 4 = Verwest (Phys. Lett. B 83, 1979) • 5 = UrQMD (Bass et al., Prog. Part. Nucl. Phys. 41, 1998)

BB_BYK		code/collisions/twoBodyReactions/baryonBaryon/barBar_BarHypKaon.f90	
enable	logical	.true.	Enable the production of BB $\rightarrow$ B Hyperon Kaon channels. B=Nucleon^{0,1},Delta^{-,0,+,,+}; Hyperon=Lambda⁰,Sigma^{0,-,+,}; Kaon=K^{+,0}

parameter_set	integer	2	Select a particular parameter set for BB→BYK collisions. Possible values: • 1 = original Tsushima model: Tsushima et al., PRC59 (1999) 369 • 2 = extended/adjusted model, fitted to HADES data: Agakishiev et al., arXiv:1404.7011 • 3 = custom parameters based on Tsushima values (as given by the array 'a' in the jobcard; those values not given in the jobcard are adopted from Tsushima, i.e. parameter set 1) • 4 = custom parameters based on HADES values (as given by the array 'a' in the jobcard; those values not given in the jobcard are adopted from HADES, i.e. parameter set 2)
a	real, dimension(1:Nch)	-1.	This array contains the "a" parameters (in microbarn) for the 30 primary channels, see: • Tsushima et al., PRC59 (1999) 369, table III • Agakishiev et al., arXiv:1404.7011, chapter IV Note: The values given in the jobcard are only used for parameter_set = 3 and 4.

Box	code/init/initBox.f90		
thermalInit	logical	.false.	flag how to initialize
nDens	real	1.0	particle density [fm ⁻³]
ChargeSelection	integer	0	define the type of the charge selection: • 0: only pi0 • 1: 50% pi+, 50% pi- • 2: 33% for +,0,-
pInit	real	0.5	initial momentum of particles [GeV/c]
BoostZ	real	0.0	additional boost for all particles in z-direction
Temp	real, dimension(1:122)	0.0	for thermal init: temperature of every meson species in GeV, if larger than 0. otherwise this species is not initialized
Fugacity	real, dimension(1:122)	1.0	for thermal init: fugacity of every hadron species.

correctMovingBox	integer	-2	switch to indicate, whether a correction of the momenta after initialization should be done to enforce vanishing 3-momenta. possibilities are: • -2: set to 3,4 according inputGeneral::fullEnsemble • -1: set to 1,2 according inputGeneral::fullEnsemble • 0 : no correction • 1 : global correction (rescaling) • 2 : per ensemble correction (rescaling) • 3 : global correction (rotating) • 4 : per ensemble correction (rotating)
noAnti	logical	.false.	if .true., no antiparticles will be initialized in thermal init
coldMatter	logical	.false.	if .true., matter at T=0 will be initialised according rhoP and rhoN
rhoP	real	-99.9	proton density for cold matter
rhoN	real	-99.0	neutron density for cold matter
useOffShell	logical	.false.	initialize cold matter box with offshell nucleons

BoxAnalysis	code/analysis/BoxAnalysis.f90		
do_Tmunu	logical	.false.	Switch for Tmunu output. default: Only one file for all ensemble! you may change this with the flag perEnsemble_Tmunu
do_TmunuZ	logical	.false.	switch for Tmunu for every z-coordinate NOTES This is mutually exclusive with perEnsemble_Tmunu
perEnsemble_Tmunu	logical	.false.	Switch for Tmunu output. One file per ensemble! NOTES this may slow down the execution dramatically, since huge output to the hard drive is induced. You may observe this, if e.g the cpu load drops permanently to 30%. Thus: switch it on, only if you want it! NOTES This is mutually exclusive with do_TmunuZ
selectTmunuFormat	integer	2	select output format of Tmunu (binary encoded): • 1: ASCII • 2: Binary • 3: ASCII + Binary
do_P	logical	.false.	Switch for dN/p^2 dp output
do_velrel	logical	.false.	Switch for calculating velrel

do_posrel	logical	.false.	Switch for calculating relative distance
do_Cumulants	logical	.false.	Switch for calculating cumulants
useSet	logical, dimen- sion(nSet)	(/ .false., .false., .false., .true., .true. /)	Array to indicate, which particle set will be used for output
factorSubBoxVolume	real	0.5	Volume of the sub box relative to the full box
do_DumpPartVec	logical	.false.	Switch for writing the whole particle vector to file at the end of the run

Checks	code/run/checks.f90		
Do_CheckDensity	logical	.false.	Flag to indicate whether the density check routine should be called.
Do_CheckCoulomb	logical	.false.	Flag to indicate whether the Coulomb check routine should be called.
Do_CheckFermiSurface	logical	.false.	Flag to indicate whether the Fermi-surface check routine should be called.
Do_CheckRadius	logical	.false.	Flag to indicate whether the radius check routine should be called.
Do_CheckMomentumDensity	logical	.false.	Flag to indicate whether the momentum-density check routine should be called.
Do_CheckEnergyLDA	logical	.false.	Flag to indicate whether the local density approximation check routine should be called.
Do_OccupiedReal	logical	.false.	Flag to indicate whether the occupation check routine for the real particle vector should be called.
Do_OccupiedPert	logical	.false.	Flag to indicate whether the occupation check routine for the perturbative particle vector should be called.
Do_CheckEnergy	logical	.false.	Flag to indicate whether the energy check routine should be called.
Do_TachyonsReal	logical	.false.	Flag to indicate whether the tachyon check routine for the real particle vector should be called.
Do_TachyonsPert	logical	.false.	Flag to indicate whether the tachyon check routine for the perturbative particle vector should be called.
TachyonIsBlocking	logical	.false.	Select whether the occurrence of a 'tachyon' in the check routines will stop the code or not (error messages will hopefully occur later in the code).
Do_CheckPertFlag	logical	.true.	Flag to indicate whether the flag '%pert' is set correctly in the particle vectors
Do_CheckConservation	logical	.false.	Flag to indicate whether conservation of energy/momentum, baryon number and strangeness between time steps for the real particles should be checked

coll_BaB code/collisions/twoBodyReactions/HiEnergy/DoColl_BaB.f90			
iset	integer	1	Switch to choose an initialization of jets: • 1: phase space distribution, also the charge is conserved (new prescription) • 2: first jet along inPart(1) momentum, 3-d jet opposite, others orthogonal, charge is not conserved (old prescription)

coll_Manni code/collisions/twoBodyReactions/HiEnergy/DoColl_Manni.f90			
angDistribution	integer	2	Switch to select the angular distribution: • 1: isotropic • 2: diquark/quark aligned like baryon/meson
itry_max	integer	10	maximum number of tries

collCriteria code/collisions/twoBodyReactions/collisionCriteria.f90			
kodama_evalFrame	logical	.false.	Set to .true., this logical will cause the kodama criterion to be evaluated in the laboratory/evaluation frame, not CM frame.

collHistory code/storage/CollHistory.f90			
DoCollHistory	logical	.false.	Flag to switch on/off the whole Collision History machinery. You may set this variable via your jobcard, namelist "collHistory"

collisionterm code/collisions/collisionTerm.f90			
oneBodyProcesses	logical	.true.	Switch on/off one-body-induced processes.
twoBodyProcesses	logical	.true.	Switch on/off two-body-induced processes.
threeBodyProcesses	logical	.true.	Switch on/off three-body-induced processes.
threeMesonProcesses	logical	.false.	Switch on/off three-meson-induced processes. These are the backreactions for e.g. $\omega \rightarrow \pi \pi \pi$ etc.
threeBarMesProcesses	logical	.false.	Switch on/off baryon-meson-meson induced processes. These are the backreactions for e.g. $N \pi \rightarrow N \pi \pi$ etc.
twoPlusOneBodyProcesses	logical	.false.	Switch on/off 2+1 body processes (two really colliding particles plus one nearby).

twoBodyProcessesRealReal	logical	.true.	Switch on/off two-body-induced processes between two real particles.
twoBodyProcessesRealPert	logical	.true.	Switch on/off two-body-induced processes between a real and a perturbative particle.
oneBodyAdditional	logical	.true.	Switch on/off additional Pythia one-body-induced processes.
doForceDecay	logical	.true.	switch on/off the forced decays at the end of the simulation NOTES • Do not touch, unless you know what you are doing! • You may set this to .false., if you are e.g. running box calculations with excited states. Decaying all these particles would need a much larger particle vector...
energyCheck	real	0.01	Precision of energy check for each collision in GeV.
maxOut	integer	100	Maximal number of produced particles in one process. Must not be smaller than 4. NOTES • When using annihilation, must not be smaller than 6 • If using FRITIOF or PYTHIA, should probably larger than 6 • Code stops with an error message, if value chosen too small
collisionProtocol	logical	.false.	Write a protocol of all real-real collisions to the file 'fort.990'. Includes the time, IDs, charges, invariant masses and 3-momenta of both collision partners and an indicator for Pauli blocking.
printPositions	logical	.false.	Switch on/off output of positions in real-pert collisions. Produces statistical output.
useStatistics	logical	.false.	Generate statistical information using the module statistics.
noNucNuc	logical	.false.	Switch on/off perturbative NN reactions.
storeRho0Info	logical	.false.	Flag whether in a rho0 decay the particle numbers of the resulting charged pions are stored or not.
storeRho0InfoOnlyDifr	logical	.false.	Flag, whether the flag storeRho0Info is valid for all decays or only for rho0, which are marked to be diffractive.
DoJustAbsorptive	logical	.false.	If this flag is true, then: for perturbative simulations all final state particles in a collision are set to zero; for real simulations %event index of incoming hadron is changed in the case of collision, but actual collision is not simulated. This is a way to mimick Glauber like calculations. NOTES The "absorption" is done with sigmaTot, not just by sigmaInEl.
annihilate	logical	.false.	If this flag is true, then an annihilation of the antibaryons with the closest baryons will be simulated (by hand) starting from annihilationTime.

annihilationTime	real	1000.	Time moment (in fm/c) when the annihilation will be started. NOTES This flag has an influence only when annihilate = .true. Before annihilationTime all the collision processes are not activated. They start to act (if the corresponding switches oneBodyProcesses,twoBodyProcesses etc. are .true.) only after annihilationTime.
justDeleteDelta	logical	.false.	Deletes final-state products in Delta N N $\rightarrow$ NNN and Delta N $\rightarrow$ N N. NOTE: Only for testing and comparing with the old Effenberger code! DO NOT USE OTHERWISE: Violates energy conservation! This switch is meant to simulate the treatment of the Delta in the old code. Only implemented for perturbative runs.
noRecollisions	logical	.false.	Outgoing particles of collisions are inserted somewhere in the particle vector. Due to implementation issues, these outgoing particles may interact during the same timestep. Setting this flag to true, the parameter '%lastCollTime' is checked against the actual time variable and collisions of these particles are excluded.

collReporter	code/collisions/collisionReporter.f90		
UseCollReporter	logical	.false.	Enable or disable the collision reporter.
cR_sizeT	integer	200	Number of timestep bins.
cR_sizeE	integer	100	Number of sqrt(s) bins.
cR_DeltaT	real	0.1	Size of timestep bins.
cR_DeltaE	real	0.1	Size of sqrt(s) bins.

ColStat	code/collisions/twoBodyReactions/twoBodyStatistics.f90		
flag_sqrts	logical	.false.	If .true., then the calculation and output of the sqrts distributions from subroutine sqrts_distribution will be done
flag_rate	logical	.false.	If .true., then the calculation and output of the collision rates from subroutine rate will be done
flag_varirate	logical	.false.	If .true., then the calculation and output of the collision rates from subroutine varirate will be done

sqrts_mode	integer	1	This flag determines the way how sqrt(s) is calculated (if flag_sqrts = .true.): • 1 = use vacuum sqrt(s) • 2 = use in-medium, i.e. full sqrts
varirate_chargeZero	logical	.true.	If .true., then all charge states are combined together
varirate_size	integer	100	size of array to hold all rates
varirate_filterPhi	logical	.false.	If .true., then only channels involving a phi meson are reported
varirate_format	integer	1	indicate the format to produce the output. It is binary coded: • 1 : use human readable output • 2 : use machine readable output (csv format) So valid values are 1,2,3.

coulomb	code/potential/coulomb/coulomb.f90		
coulombFlag	logical	.false.	Switch to turn on/off the Coulomb potential. If turned on, also 'symmetryPot-Flag' (in namelist 'baryonPotential') needs to be turned on.
magnetFieldFlag	logical	.false.	Switch to turn on/off elm. vector potential. NOTES The vector potential is not yet fully implemented! Please do not use this option.
cutMomentumPotential	real	0.025	If larger than 0, the coulomb potential is set to zero for all particles with momentum larger than $\text{minmass}^2/(2*\text{cutMomentumPotential})$ The cut-off is smeared out, if cutMomentumWidth>0 This cut is needed in non-RMF mode since $i\text{pre}_i m_{\text{eff}}^2 = (\text{sqrt}(m^2+p^2)+U_C)^2-p^2$ can become smaller than zero for $i\text{pre}_i p > U_C/2 - m^2/(2*U_C)$. In this case we have a 'tachyon'. NOTES • for RMF mode you do not need this cut • This value should correspond to the maximum vale of the Coulomb potential. Therefore you should readjust this for every nucleus. • For the pion, we take the mass (0.138) instead of minmass, since here minmass is zero!

cutMomentumWidth	real	0.100	If cutMomentumPotential>0 and cutMomentumWidth>0, then the cut-off is smeared by some linear interpolation: $\text{pre}_i(\text{At-Width}) = 1.0 \dots (\text{At+Width}) = 0.0$ with $\text{At} = \min\text{-mass}^2/(2*\text{cutMomentumPotential})$ and the width given here in GeV. This width is introduced in order to destroy numerical problems due to sharp edges.
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DecayChannels

code/database/decayChannels.f90

rhoDelta_is_sigmaDelta	logical	.false.	If true, the rho-Delta decay channel will be replaced by sigma-Delta. For discussion, see e.g. Effenberger PhD, chapter 6.3.2.
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deltawidth

code/width/deltaWidth.f90

deltaSwitch	integer	3	Switch for different prescriptions for the delta width. NOTES • 1 = Oset self energies+BUU input • 2 = Spreading potential • 3 = Oset self energies • 4 = density dependent with BUU input
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detailed_diff

code/analysis/neutrinoAnalysis.f90

EkinMin	real	0.	if detailed_diff_output is TRUE: minimal kinetic energy for dsigma/dEkin for hadrons
EkinMax	real	2.	if detailed_diff_output is TRUE: Maximal kinetic energy for dsigma/dEkin for hadrons
dEkin	real	0.01	if detailed_diff_output is TRUE: Delta(eKin) for dsigma/dEkin for hadrons
forPion	logical	.true.	If .true. then also the detailed output of differential cross sections is produced
forEta	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forKaon	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forKaonBar	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forDmeson	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forDbar	logical	.false.	If .true. then also the detailed output of differential cross sections is produced

forDs_plus	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forDs_minus	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forNucleon	logical	.true.	If .true. then also the detailed output of differential cross sections is produced
forLambda	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forSigmaResonance	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forXi	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
forOmegaResonance	logical	.false.	If .true. then also the detailed output of differential cross sections is produced

deuteriumFermi code/init/deuterium.f90			
waveFunction_switch	integer	1	Possible values are: • 0 – No Wave functions! Pointlike Deuterium • 1 – Wave functions according to Bonn potential • 2 – Wave functions according to Argonne V18
iParam	integer	1	Choose parameterization of momentum distribution when using the Bonn potential. Possible values: • 1 – Full Bonn (MaH87) • 2 – OBEPQ (MaH87) • 3 – OBEPQ-A (Mac89) • 4 – OBEPQ-B (Mac89) • 5 – OBEPQ-! (Mac89) • 6 – OBEPR (MaH87) self-made • 7 – Paris References: • MaH87: R. Machleidt et al. Phys. Rep. 149, 1 (1987) • Mac89: R. Machleidt, Advances in Nucl. Phys. Vol 19
pMax	real	0.5	Cut-off parameter for Fermi momentum
scaleMomentum	real	1.0	The selected momentum is multiplied by this factor afterwards, i.e. some rescaling is done

DileptonAnalysis code/analysis/DileptonAnalysis.f90			
Enable	logical	.false.	If .true. the dilepton analysis will be performed, otherwise not.

Extra	logical	.false.	If .true. an extended analysis will be performed, writing out many extra histograms (beyond the basic ones: mass, pT and rapidity).
DeltaDalitz	integer	2	Choose between different parametrizations of the Delta Dalitz decay width ($\Delta \rightarrow N e^+e^-$): • 1 = Wolf, http://inspirehep.net/record/306273 • 2 = Krivoruchenko (default), http://inspirehep.net/record/555421 • 3 = HadronTensor • 4 = Ernst, http://inspirehep.net/record/452782
DeltaDalitzFF	integer	1	Choose a parametrization of the electromagnetic N-Delta transition form factor for the Delta Dalitz decay (only used for DeltaDalitz = 2): • 1 = constant (default) • 2 = Dipole • 3 = MAID 2005 • 4 = simple VMD • 5 = Wan/Iachello, Int. J. Mod. Phys. A 20 (2005) 1846, http://inspirehep.net/record/689265 • 6 = Ramalho/Pena, Phys.Rev. D85 (2012) 113014, http://inspirehep.net/record/1114321
omegaDalitzFF	integer	1	Choose between different parametrizations of the omega Dalitz decay ($\omega \rightarrow \pi^0 e^+e^-$) form factor: • 0 = constant • 1 = Effenberger/Bratkovskaya (default) • 2 = standard VMD • 3 = Terschluesen/Leupold
b_pi	real	5.5	This constant represents the b parameter in the form factor of the π^0 Dalitz decay (in GeV^{-2}), cf. Effenberger Diss. eq. (2.141). Originally taken from L.G. Landsberg, Phys. Rep. 128, 301 (1985).
lambda_eta	real	0.716	This constant represents the Lambda parameter in the form factor of the eta Dalitz decay in GeV. Values: • L.G. Landsberg, Phys. Rep. 128, 301 (1985): $\Lambda = 720 \text{ MeV}$ • HADES pp@2.2, B. Spruck, Diss. (2008): $\Lambda = 676 \text{ MeV}$ • NA60, Arnaldi et al, PLB 677 (2009): $\Lambda = 716 \text{ MeV}$ (default) • CB/TAPS, Berghäuser et al, PLB 701 (2011): $\Lambda = 722 \text{ MeV}$

etaPrimeDalitzFF	integer	0	Choose between different parametrizations of the eta' Dalitz decay ($\eta' \rightarrow e^+e^- \gamma$) form factor: • 0 = constant (default) • 1 = eta FF (cf. lambda_eta) • 2 = generic VMD • 3 = Genesis / Lepton-G • 4 = standard VMD (Terschluesen)
angDist	integer	1	This switch determines the angular distribution of the pseudoscalar Dalitz decays $P \rightarrow e^+ e^- \gamma$ (with $P=\pi^0, \eta, \eta'$): • 0 = isotropic decay • 1 = anisotropic decay according to $1 + B \cos^2(\theta)$ with $B=1$ • 2 = the Dalitz decays of π^0 and η will be done via Pythia.
brems	integer	1	This switch determines how the bremsstrahlung contribution is obtained: • 0 = none • 1 = soft-photon approximation (SPA) • 2 = according to the one-boson-exchange (OBE) model by R. Shyam (for NN bremsstrahlung only, no em. form factors) • 3 = as 2, but with pion em. form factor (for pn) • 4 = as 3, but times correction factor (for pn), see arXiv:2009.11702
nEvent	integer	10	Number of events to generate for each dilepton decay (to enhance statistics).
nEvent_BH	integer	1000	Number of events for Bethe-Heitler simulation. BH typically needs a lot more statistics than the other dilepton channels. Therefore nEvent_BH should be much bigger than nEvent.
kp_cut	logical	.false.	Perform a cut on $(k \cdot p)$ in the dilepton analysis, where k is the photon 4-momentum, and p is the electron or positron 4-momentum. This is useful for suppressing the BH contribution. Cf. "kp_min".
kp_min	real	0.01	If kp_cut=.true. a cut on $(k \cdot p)$ is performed. kp_min determines the position of this cut. Only events with $(k \cdot p) > kp_min$ are taken into account.
binsz	real	0.01	This determines the bin size of the dilepton mass spectrum in GeV. Default is 10 MeV.

filter	integer	0	If filter is nonzero, a filtering algorithm will be applied to the dilepton pairs, otherwise they will be written to the histograms unfiltered. For details on the filtering parameters see routine 'CS'. Choices: • 0 = no filter • 1 = DLS • 2 (removed) • 3 = HADES (full acceptance filter, using pair acceptance) • 4 = HADES (full acceptance filter, using single-particle acceptance) • 5 = g7/CLAS @ JLab • 6 = KEK E325 (cuts on rapidity, transverse momentum and opening angle) • 7 = JPARC E16 • 10 = HADES (simple cuts on polar angle, abs. momentum and opening angle) • 11 = HADES (as 10, but modified by Jan Otto for AgAg@1.58) • 12 = HADES (as 10, but modified by Karina Scharmann, pp@1.58) • 20 = as 10, but with smearing (needs dummy HAFT file) • 21 = as 11, but with smearing (needs dummy HAFT file) • 22 = as 12, but with smearing (needs dummy HAFT file) • 30 = as 10, but with smearing according SmearFile • 31 = as 11, but with smearing according SmearFile • 32 = as 12, but with smearing according SmearFile NOTES For filtering modes 3 and 4, the file containing the acceptance matrices must be specified (cf. <code>hadesFilterFile</code>). The old filter 2 has been renamed to 10
hadesFilterFile	character(len=1000)		This character string determines the location of the file containing the HADES acceptance matrices (filename with absolute or relative path). For filter=4, some default files are selected, if this input parameter is left empty. possible values: • if not set, default is '[path.To.Input]/hades/XXX' (for filter=4) • if given, but does not contain '/': default is '[path.To.Input]/hades/[hadesFilterFile]' • otherwise: filename is absolute, including path NOTE if you want to use the file 'XXX.dat' in the actual directory, give it as './XXX.dat'

hadesSmearFile	character(len=1000)		This character string determines the location of the file containing the HADES smearing matrices (filename with absolute or relative path). for filter=30,31,32, a value must be given: • if given, but does not contain '/': default is '[path_To_Input]/hades/[hadesSmearFile]' • otherwise: filename is absolute, including path NOTE if you want to use the file 'XXX.dat' in the actual directory, give it as './XXX.dat'
WriteEvents	integer	0	This switch decides whether we write out the simulated events. Possible values: • 0: Don't write events (default). • 1: We write out only the lepton pair information (including charge, four-momentum, perturbative weight, source channel and filter result). All of this will be written to a file called 'Dilepton_Events.dat'. • 2: As 1, but only writing exclusive events ($NN \rightarrow NNe+e-$). • 3: We write out all produced particles in the event (including the lepton pair) to a file called 'Dilepton_FullEvents.dat'. • 4: As 3, but only writing exclusive events ($NN \rightarrow NNe+e-$). • 5: As 4, but only writing out $R \rightarrow Ne+e-$ events (with $R=N^*, \Delta^*$).
p_lep_min	real	0.	This switch sets a lower bound on the lepton momentum. Only leptons with momenta larger than this threshold will pass the filter. This switch is only used for filtering mode 5 (JLab).
beta_gamma_cut	real	1.25	This is an upper bound on the $\beta^*\gamma$ value of the lepton pair. Since $\beta^*\gamma = p/m$, it cuts on slow pairs.
massBinning	real, dimension(1:4)	(/ 0.150, 0.550, 9.999, 9.999 /)	We produce several histograms (e.g. $p, p_T, m_T, y, \theta_{cm}$) not only for the full mass range, but also for (up to 5) different mass bins. The borders of these bins are given by this array.
particle_source	logical	.true.	This switch determines whether the mass spectrum will contain separate channels for different sources of particles, such as decays ($R \rightarrow \rho N$) or collisions ($\pi\pi \rightarrow \rho, KK \rightarrow \phi$). Currently this is only done for the ρ and ϕ mesons. Note: If using this switch, the "sum" channel in the mass histogram should not be used, since the ρ and ϕ contributions will enter twice.
missMass_min	real	-99.9	if > 0, only events with a missing mass larger than this value are taken into account
missMass_max	real	-99.9	if > 0, only events with a missing mass smaller than this value are taken into account
JanOttoOmega	logical	.false.	If .true. an extended analysis will be performed, writing out dilepton spectra focussing on the omega region

elementary code/init/initElementary.f90			
impactParameter	real	-1.	• ≥ 0: this is the actual impact parameter • < 0: Impact parameter integration up to an automatically determined <code>b_max</code>. The actual impact parameter is randomly sampled in the interval $[0, b_max]$ with a proper geometrical weight.
particleId	integer, dimension(2)	(/1,1/)	Id of particles
particleAnti	logical, dimension(2)	(/.false.,.false./)	if .true. then particles are antiparticles
particleCharge	integer, dimension(2)	(/0,0/)	Charge of the particles
srtsRaiseFlag	logical	.false.	• if .true. then the srts stepping is done • if .false. then the <code>ekin_lab</code> stepping is done
ekin_lab	real	1.	kin. energy of first particle in the rest frame of second particle (starting value for the energy scan: the number of different energies is set by parameter <code>num_Energies</code> in the namelist "input")
delta_ekin_lab	real	0.03	kin. energy step for energy scan
srts	real	3.	invariant energy (starting value for the energy scan)
delta_srts	real	1.	srts step for srts scan

Elementary_Analysis code/analysis/ElementaryAnalysis.f90			
DoOutChannels	logical	.false.	switch on/off: reporting of all final state channels
DoH2d	logical	.false.	if .true. than make output of 2-dimensional histograms (they could produce files of size 240 mb)
Do45ForAllEvents	logical	.false.	flag to decide, whether <code>DoElementaryAnalysis4(5).dat</code> is written for all events or just for events, where the output channel consist of pions
DodNNbar	logical	.false.	
DoPanda	logical	.false.	if .true., elementary analysis for channels with $S = -2$ and -3 (Ξ , Ω)
Dodsigdt	logical	.false.	
Do2Part	logical	.false.	

eN_Event code/init/ElectronGenerator/eN_event.f90			
selectFrame	integer	2	select frame, in which the calculaton of W_free is done: • 0 = doNOT — do NOT remove • 1 = CM • 2 = CALC • 3 = THRE prescription from correct threshold behaviour, used in heavy ion collisions • 4 = NucleonRest : boost nucleon in the rest frame, set free mN, recalculate boson momentum • 5 = THRE2 threshold with m^2: $s_{\text{free}} = s + m^2 - m^{*2}$
restingNucleon	logical	.true.	if this flag is .false., we use the momentum of the target nucleon in the calculation of the flux

EventOutput code/analysis/EventOutputAnalysis.f90			
WritePerturbativeParticles	logical	.false.	Flag to write out the perturbative particle vector to an output file. The switch 'EventFormat' determines which format is used.
WriteRealParticles	logical	.false.	Flag to write out the real particle vector to an output file. The switch 'EventFormat' determines which format is used.

EventFormat	integer	1	This switch determines the format of the event output files. Possible values: • 1 = Les Houches format (default) • 2 = OSCAR 2013 format • 3 = Shanghai 2014 format • 4 = ROOT • 5 = OSCAR 2013 extended format • 6 = HepMC3 (asciiv3, via library) • 7 = NuHepMC (asciiv3) NOTES • For Les Houches, the output will be written to files called EventOutput.Pert.lhe and EventOutput.Real.lhe. • For OSCAR, the output files are called EventOutput.Pert.oscar and EventOutput.Real.oscar. • For Shanghai, the output files are called EventOutput.Pert.dat and EventOutput.Real.dat. • For ROOT, the output files are called EventOutput.Pert.root and EventOutput.Real.root. • For HepMC, the output files are called EventOutput.Pert.hepmc3 and EventOutput.Real.hepmc3. • For NuHepMC, the output files are called EventOutput.Pert.hepmc3 and EventOutput.Real.hepmc3.
Interval	integer	0	Interval for event output, i.e. number of timesteps after which output is written. If zero, only final output at the end of the time evolution is produced.

externalSystem code/init/initExternal.f90			
inputFile	character*1000	'./source.inp'	the absolute name of the input file with hadrons to be propagated. possible values: • if not set, default is './source.inp' • if given, but does not contain '/': default is './[inputFile]' • otherwise: filename is absolute, including path NOTE if you want to use the file 'XXX.inp' in the actual directory, give it as './XXX.inp'
DoPerturbative	logical	.false.	if true, the particles will be inserted into the perturbative particle vector, the real particles have to be initialized via some nucleus definition

NumberingScheme	integer	1	The way, how particles%event will be numbered: • 1: event = iPart, i.e. the particle number in the ensemble (historical, but does not work for fullensemble) • 2: event = pert_numbering() or real_numbering() (good both for perturbative and real mode)
posSRC	logical	.false.	If true, the position vectors of the proton and neutron from SRC will be sampled by Monte-Carlo. Relevant when the target nucleus was initialized before calling initializeExternal and if there are only proton and neutron in the external source.
flagPH	logical	.false.	If true, a particle-hole excitation will be added to the target nucleus. The momentum of the particle is obtained by adding transverse momentum transfer along x-axis and from "–" momentum conservation.
pt	real	0.	Transverse momentum transfer to the struck nucleon (GeV/c). Relevant when flagPH=.true.
ff_QE	code/init/lepton/formfactors_QE_nucleon/FF_QE_nucleonScattering.f90		
parametrization	integer	3	• 0 = dipole approximation • 1 = BBA03 parametrization • 2 = BBBA05 parametrization • 3 = BBBA07 parametrization
MV2	real	0.71	vector mass squared in the dipole parametrization of the vector form factors
MA_in	real	1.0	axial mass (only if useNonStandardMA=.true. and will only impact CC if parametrizationAxial=0,1).
useNonStandardMA	logical	.false.	if one wants to use a specific axial mass, set this to true and choose value for MA_in, will only impact CC if parametrizationAxial=0,1.
deltas	real	-0.15	strange contribution to the axial ff.
parametrizationAxial	integer	0	• 0 = dipole approximation • 1 = use axial ff. of Gari, Kaulfuss PLB 138 (1984) • 2 = use z-expansion from MINERvA fit in Nature 614, 48-53 (2023) • 3 = use z-expansion from LQCD fit in arXiv:2512.14097 eq (41) • 4 = use z-expansion from Deuterium fit in arXiv:2512.14097 (50) • 5 = use z-expansion with custom parameters, specified with t0,tc,a1,a2,...
kmax	integer	20	number of ak used in z-expansion (only if parametrizationAxial=5).
t0	real	-0.75	parameter used in calculating z (only if parametrizationAxial=5).

tc	real	0.173889	cutoff parameter used in calculating z (only if parametrizationAxial=5).
ak			

FinalState_Full	code/collisions/phaseSpace/finalState_Full.f90		
maxbwd_scalingFactor	real	1.	• Rescales maxBWD
silentMode	logical	.true.	• Switches error messages off in massAss. Errors can still be seen looking at massAssStatus.dat
NYK_isotropic	logical	.false.	If .true., the angular distribution in Nucleon-Hyperon-Kaon production is assumed to be isotropic. If .false., a non-isotropic distribution is used, as described in Larionov/Mosel, Phys.Rev. C 72 (2005) 014901. See also momenta.in.3Body_BYK.

formfactors_pion	code/init/lepton/formfactors_pionProduction/formfactors_A_main.f90		
which_MaidVersion	integer	2	choice of MAID version: 1=2003, 2=2007
fineGridQ2	logical	.false.	if .true. use MAID tables with a finer grid in Q^2 ($dQ^2 = 0.05\text{GeV}^2$) instead of default $dQ^2 = 0.1\text{GeV}^2$

Freezeout	code/analysis/FreezeoutAnalysis.f90		
FreezeoutAnalysis_Pert	logical	.false.	Flag to do freeze out analysis for perturbative particles
FreezeoutAnalysis_Real	logical	.false.	Flag to do freeze out analysis for real particles
potThreshold	real	0.005	threshold value in GeV. If the absolute value of the potential is below this value, the particle is considered to be 'free', e.g. it 'escaped'

gamma_2Pi_Xsections	code/init/lowPhoton/twoPi_production/gamma2Pi_Xsections.f90		
experimentalXsections	logical	.true.	• If .true. then the Xsections are taken from the experiment • If .false. then the theoretical values are given

hadron		code/init/initHadron.f90	
impactParameter	real	0.	smaller 0: Impact parameter will be chosen randomly in the interval $[0; \text{abs}(\text{impactParameter})]$ (see subroutine setGeometry). It is recommended to take very large negative value of impactParameter in order to have good automatic random choice, e.g. impactParameter=-100.
bRaiseFlag	logical	.false.	if .true.: actual impact parameter will be raised by deltaB after nRunsPerB subsequent runs. Starting value is given by the impactParameter variable.
deltaB	real	0.	impact parameter step (relevant if bRaiseFlag=.true.)
nRunsPerB	integer	1	number of subsequent runs per impact parameter (relevant if bRaiseFlag=.true.)
perturbative	logical	.false.	if .true. then the hadron is a perturbative particle
numberParticles	integer	200	Number of projectile testparticles per ensemble in the case of a perturbative treatment
particleId	integer	1	Identity of the projectile hadron.
antiParticle	logical	.false.	if .true. then the hadron is an antiparticle
particleCharge	integer	0	Charge of the hadron
ekin_lab	real	1.	kinetic energy of the hadron in the rest frame of the target nucleus (GeV) NOTES If ekin_lab < 0. — initialization according to the binding energy
E_bind	real	0.	binding energy of initialized hadron (GeV) NOTES Active for iniType= 0,2 if ekin_lab < 0. is set.
iniType	integer	0	• 0: usual initialization for the hadron-nucleus collision • 1: position and momentum of the hadron is chosen according to the Gaussians centered, resp., at the centre of the nucleus and at zero momentum (impactParameter, distance and ekin_lab have no effect in this case) • 2: gaussian in coordinate space, but usual sharp momentum choice (impactParameter, distance and ekin_lab work as usual)
zChoice	integer	1	• 1: hadron is initialised at fixed distance delta from nuclear surface • 2: hadron is initialised at fixed z Relevant for iniType=0 or iniType=2.

delta	real	0.5	• for zChoice=1: distance from nuclear surface, at which the hadron is initialised [fm] • for zChoice=2: maximum distance from the edge of nucleus in transverse direction which restricts the choice of actual impact parameter for impactParameter < 0 (for impactParameter > 0 no restriction) Relevant for iniType=0 or iniType=2.
deltaZ	real	5.	z = -deltaZ - R_nucleus, where z is z-coordinate of the hadron Relevant for iniType=0,2 and zChoice=2.
width	real	1.	Width of a gaussian density profile [fm]. Only relevant for iniType= 1 and 2.

HadronAnalysis code/analysis/hadronAnalysis.f90			
flagAnalysis	logical	.false.	If true, perform the output of a hadron at the latest time step before the hadron disappeared (file DoHadronAnalysisTime.dat). The output hadron has the same baryon/meson type and antiparticle-flag as the beam particle. In case if the hadron did not disappear, the output is done at the end of the time evolution. The output for the hadron is also done in three other files if its momentum becomes for the first time less than the cut values pCut1 and pCut2 (files DoHadronAnalysisTime1.dat and DoHadronAnalysisTime2.dat) and if it becomes bound (DoHadronAnalysisTime3.dat) NOTES Presently feasible only for real particle simulations.

hadronformation code/collisions/twoBodyReactions/hadronFormation.f90			
tauProda	real	0.5	in formation time concept 2) and 3) for "error particles": production time of non-leading in rest frame of hadron (in fm)
tauForma	real	0.8	in formation time concept 1) and in concept 2),3) for "error particles": formation time in rest frame of hadron (in fm)
tauFormaFak	real	1.0	in formation time concept 1): scale factor for constituent quark model, rescales #(lead quarks)/#quarks
useJetSetVec	logical	.true.	Flag to select fragmentation time estimates: • false → old concept 1) • true → new concepts 2) and 3) NOTES select false in case of calculations on a nucleon (speed!).

powerCS	real	1.0	in formation time concept 2): power of 't' (constant, linear, quadratic)
useTimeFrom	integer	1	in formation time concept 2): encode time XS starts to evolve: 1: tP_min, 2: tP_max, 3: tF
useTimeTo	integer	3	in formation time concept 2): encode time XS stops to evolve: 1: tP_min, 2: tP_max, 3: tF
GuessDiffrTimes	logical	.true.	if true, then the times for diffractive particles are treated like them of all other particles, otherwise particles from "diffractive" events hadronize immediately.
useJetSetVec_Q	logical	.true.	if useJetSetVec, then also use Q2 as measure for XS-pedestal, i.e. select concept 3) instead of concept 2)
useJetSetVec_R	logical	.true.	if not useJetSetVec_Q, then use rLead as measure for XS-pedestal
pedestalCS	real	0.0	in formation time concept 2): encode time XS stops to evolve: 1: tP_min, 2: tP_max, 3: tF
useQDM	logical	.false.	If true, then use the quantum diffusion model of G.R. Farrar et al., PRL 61, 686 (1988). It means that the cross section grows as $(t-t_{int})^{**powerCS}$ for $t_{int} < t < t_{form}$, where t_{int} is the interaction time (=0 for electron-nucleus case) and $t_{form} = t_{int} + 2*p/dM2$. So hadrons with equal momenta have equal formation times (lengths). Also allows to control the space-time scale of hadronization. Attention: setting useQDM = .true. overrides other switches of this module.
dM2	real	0.7	Mass denominator in the coherence length. Relevant only for quantum diffusion model (when useQDM =.true.)
use_pCut	logical	.false.	If true, then only particles with momentum $p < pCut$ will interact.
pCut	real	1.	Momentum cutoff. Relevant only when use_pCut =.true.

HadronTensor_ResProd	code/init/lepton/hadronTensor_ResProd.f90	
speedup	logical	.true.

heavyIon		code/init/initHeavyIon.f90	
impact_parameter	real	0.	Impact parameter b [fm]. There are three options: • $b \geq 0$: The impact parameter is fixed to the given value. • $-100 < b < 0$: The impact parameter will be chosen randomly in each run between 0 and $\text{abs}(b)$. • $b \leq -100$: "Minimum bias". The impact parameter will be chosen randomly in each run (maximum = sum of radii plus twice the sum of surfaces).
impact_profile	integer	0	This switch provides impact-parameter distributions for trigger-biased setups. Only used for impact_parameter < 0. Possible values: • 0: minimum bias (default) • 1: HADES C+C at 1.00 AGeV • 2: HADES C+C at 2.00 AGeV • 3: HADES Ar+KCl at 1.76 AGeV • 4: HADES Au+Au at 1.23 AGeV (all) • 5: HADES Au+Au at 1.23 AGeV (0-10% central) • 6: HADES Au+Au at 1.23 AGeV (10-20% central) • 7: HADES Au+Au at 1.23 AGeV (20-30% central) • 8: HADES Au+Au at 1.23 AGeV (30-40% central) • 9: HADES Ag+Ag at 1.58 AGeV (0-40% central) [not in release] • 10: HADES Ag+Ag at 1.58 AGeV (0-10% central) [not in release] • 11: HADES Ag+Ag at 1.58 AGeV (10-20% central) [not in release] • 12: HADES Ag+Ag at 1.58 AGeV (20-30% central) [not in release] • 13: HADES Ag+Ag at 1.58 AGeV (30-40% central) [not in release]
distance	real	0.	Distance between centers of nuclei along z (i.e. beam)-direction [fm]. This will be readjusted automatically in case it is too small.
coulomb	logical	.false.	If .true., then a Coulomb propagation from coulombDistance = 10000 fm to distance is performed.
ekin_lab_Target	real	0.	Kinetic energy per nucleon of target nucleus in lab frame [GeV].
ekin_lab_Projectile	real	0.	Kinetic energy per nucleon of projectile nucleus in lab frame [GeV].
adjustGridFlag	logical	.false.	If .true., the grid spacing in z-direction will be readjusted.
cmsFlag	logical	.true.	If .true., the collision takes place in the CM frame of the two nuclei (default option). If .false., the collision takes place in the LAB frame (target at rest).

HICanalysis_Input code/analysis/HeavyIonAnalysis.f90			
flag_outputReal	logical	.false.	If .true., then the output of the real particle vector will be written to the file 'DoHIA.dat'.
flag_outputPert	logical	.false.	If .false., then the output of the perturbative particle vector will be written to the file 'DoHIA_pert.dat'.
flag_outputDetailed	logical	.false.	Print out more detailed information at each time step from subroutine HeavyIon_evol: • rhorad_*.dat • rhoz_*.dat • rhozx_*.dat • Fields_*.dat • pauli_*.dat • dens_max.dat
pionAnalysis	logical	.false.	This flag generates various pion spectra (p-T, m-T, y, etc). The analysis operates under the assumption of a fixed target, and expects the collision to be performed in the CMS system (cf. cmsFlag in namelist /heavyIon/). The analysis matches the one applied to the HADES data in Agakishiev et al., Eur.Phys.J. A40 (2009) 45-49.
etaAnalysis	logical	.false.	This flag generates various eta spectra and eta-related analyses.
rapBinningMode	integer	1	Select the variable the 'rapBinning' is given for: • 1: variable is $y_0 = y/y_{\text{cms}}$ (the normalised rapidity) • 2: variable is y
rapBinning	real, dimension(0:13)	(/ -0.75, -0.45, -0.15, 0.15, 0.45, 0.75, 1.05, 1.35, -99.9, -99.9, -99.9, -99.9, -99.9 /)	Rapidity binning for the pion and eta analysis (only used if pionAnalysis = .true. or etaAnalysis = .true.). The numbers represent the binning borders in y (or y0, see rapBinningMode). For each of the bins, a separate pT and/or mT spectrum will be generated. Only bins, where the upper bound is larger than the lower one are considered NOTES • for the Hades AuAu analysis, you should set the bins to -0.65,-0.55,...0.75
KaonAnalysis	logical	.false.	This flag generates various Kaon spectra and Kaon-related analyses.
DensityPlot	logical	.false.	This flag select printing the density for several time steps
NucleonMassPlot	logical	.false.	This flag select printing the (invariant) mass of the nucleons for several time steps
do_Spectra	logical	.false.	Switch for general spectra output.
do_QRvector	logical	.false.	Switch for QRvector output.

do_Glauber	logical	.false.	Switch for Glauber-MC analysis at timestep 0
do_Tmunu	logical	.false.	Switch for Tmunu output.
BarMes_Tmunu	logical	.false.	If .true., then Tmunu is calculated for baryons and mesons separately.
rotateZ_Tmunu	logical	.false.	select, whether the particles are first rotated to be aligned to the z-axis
correctPot_Tmunu	integer	0	select, whether the energy is corrected for the potential or not: • 0: no correction • 1: full potential added to p0 • 2: only $U_b/2 + U_r$ added to p0 • 3: $U_b/2 + U_r$ added to p0 in the LRF
selectTmunuFormat	integer	2	select output format of Tmunu (binary encoded): • 1: ASCII • 2: Binary • 3: ASCII + Binary
useSet	logical, dimension(nSet)	(/ .false., .true., .false., .true., .true.)	Array to indicate, which particle set will be used for output
nPartAnalysis	logical	.false.	This flag generates output about impact parameter and N_part

HiGammaNucleus	code/init/ElectronGenerator/eventGenerator.eN.HiEnergy.f90		
DoLowEv	logical	.true.	If this flag is set true, then for $W_{free} < HighEnergyThreshold$ we will call the low energy model routines.
DoTransEv	logical	.false.	flag: use transitionEvent in order to replace PYTHIA events by events where we give the cross section explicitly and do the remaining stuff by FRITIOF NOTES this replaces the flag "FRITIOF" in the namelists "HiLeptonNucleus" and "HiPhotonNucleus"
useHermesPythiaPars	logical	.false.	flag: Use "PYTHIA tuning done by HERMES collab"
DoDiff	logical	.true.	flag: Generate diffractive events
PYTHIAthresh	real	2.0	Below this value for W, PYTHIA is not used to generate (G)VMD events NOTES This value is transferred to PyVP.f. you can access this value by the function "GetPYTHIAthresh()".

useVMD_VM	logical, dimension(4)	.true.	These flags can be used to switch on/off some VM in the VMD description of the events generated by "transitionevent" NOTES • The VMD events of PYTHIA are not affected. (We could change this!)
useRes	logical, dimension(2:nres+1)	.true.	Switch for including/excluding specific resonances
allowRes	logical	.true.	Switch for including/excluding resonance contribution. If this is set to .true., 1pion events will just be generated as for the background, but according the full MAID cross section (if at all)
allow1pi	logical	.true.	Switch for including/excluding 1pion contribution. Depending on the switch allowRes, 1 pion events will be done according the full cross section or just as a background.
allow2piBack	logical	.true.	Switch for including/excluding additional 2pion background.
allowDIS	logical	.true.	Switch for including/excluding DIS contribution
allowVMDrho	logical	.true.	Switch for including/excluding the VMD gamma N $\rightarrow$ rho0 N contribution
DoToyModel_pi	logical	.false.	flag: Use a Toy model instead of realistic event generation. Only a single pion is generated
DoToyModel_rho	logical	.false.	flag: Use a Toy model instead of realistic event generation Only rho0 N events are generated. Additional assumptions: (c.f.UseFormTime_ToyModel_rho) * tau_F = 0 * tau_F = m with t_F = E (boost according E/m, not E_string/M_string) In the latter case we suffer also the following simplifications: • no Q2 dependance • XS starts with n_L/n = 0.5 (should be 0.66 for the nucleon)
UseFormTime_ToyModel_rho	logical	.false.	flag: if .true., we set the formation times of the particles produced in the Toy-Model_rho equals to the energy of the particle (t_f/fm=E/GeV) representing the assumption tau_f/fm = m/GeV plus a boost according E/m. (Otherwise the formation time is set to zero.)
DoExclPiModel	logical	.false.	flag: Use a model for exclusive pion production. Only those events are generated
ExclPiCharge	integer	1	variable to specify the charge of the pion produced, if DoExclPiModel is selected
flagTwoJets	logical	.false.	If .true. - the events without two jets with large transverse momentum are marked with XS_tot=100000 mub.

HiLepton_Analysis code/analysis/HiLeptonAnalysis.f90			
DoTimes	logical	.false.	switch on/off: reporting of times
DoOutChannels	logical	.false.	switch on/off: reporting of all final state channels
DoInvMasses	logical	.false.	switch on/off: reporting of pairwise-invariant-masses
DoFindRho0	logical	.false.	switch on/off: reconstructing rho0 from final pions
DoClasie	logical	.false.	switch on/off: Do pion analysis as Clasie et al., arXiv:0701.1481
DoMorrow	logical	.false.	switch on/off: Do pion analysis as Morrow et al., Morrow:2008ek
DoBrooks	logical	.false.	switch on/off: Do pi+ pT2 spectra for Brooks delta pT2
DoMandelT	logical	.false.	switch on/off: Do pion analysis with Mandelstam t.
DoClassifyFirst	logical	.false.	Classifying 'FirstEvent' into some classes Needs DoEventAdd.
DoFSIsqrts	logical	.false.	switch on/off: Estimate potential/future final state interactions Plot the sqrt(s) distribution of potential final state interactions of perturbative particles with the nucleus (real) particles). (The interactions do not happen, this is calculated before every propagation.) In order to select the particle class for which one wants to report the FSI, please change directly the code.
DoCentralN	logical	.false.	switch on/off: Do centrality analysis with slow nucleons
DoLeptonKinematics	logical	.false.	switch on/off lepton kinematics output
DoHadronKinematics	logical	.false.	switch on/off hadron kinematics output
flagDoIt	logical	.true.	switch on/off using DoHiLeptonAnalysis

HiLeptonNucleus code/init/initHiLepton.f90			
iExperiment	integer	0	choice of experiment, detector and energy possible values are: • 0: no experiment/fixed kinematics • 1: Hermes, 27GeV, D,N,Kr • 2: Hermes, 27GeV, Ne • 3: Hermes, 27GeV, H • 4: JLAB, 12GeV • 5: JLAB, 5GeV • 6: EMC, 100GeV • 7: EMC, 120GeV • 8: EMC, 200GeV • 9: EMC, 280GeV • 10: Hermes, 12GeV • 11: Hermes, 27GeV, arXiv:0704.3270 • 12: Mainz, Yoon: Ebeam=1.5GeV • 13: Hermes, 27GeV, arXiv:0704.3712 (pT-broadening) • 14: JLAB, 5GeV, rho0 experiment • 15: JLAB, 4GeV, rho0 experiment • 16: EIC, E_e and E_A given explicit (3+30,11+30,4+100) • 17: no detector, total cross section, Ebeam • 18: E665, 470GeV • 19: CLAS/JLAB, 12GeV RunGroupA optimized 10.6 GeV • 20: CLAS/JLAB, 12GeV RunGroupA theoterical please note: The entry "iExperiment == 0" replaces the old HiPhoton event type.
shadow	logical	.true.	flag: Consider shadowing or not
minimumMomentum	real	0.1	minimal momentum considered. (in GeV)
ModusCalcFluxNorm	logical	.false.	if this flag is true, than we do not really generate events. We only select nu and Q2 according an equal distribution and plot the flux (and the flux multiplied with AccWeight). Normally we choose nu and Q2 according flux*Accweight via von-Neumann- rejection method (where we loose access to the absolute normalisation).

iDetector	integer	-1	This sets the treatment of the detector: • -1 : not valid/not initialized/use default • 0 : no detector, as AccFlag=.false. • 1 : HERMES, full efficiency • 2 : EMC, full efficiency • 3 : CLAS, only cuts (th_e=12°..50°, th_hadron=6°..143°) • 4 : CLAS, full efficiency + cuts as for 5GeV • 5 : CLAS, electron: cuts (th_e=12°..50°), hadrons: efficiency+cuts as for 5GeV • 90 : full acceptance
EIC_Ee	real	-99.9	the electron beam energy, if iExperiment=EIC
EIC_EA	real	-99.9	the hadron beam energy, if iExperiment=EIC
realRun	logical	.false.	Flag to indicate, whether we produce real or perturbative particles. NOTES run with real particles untested !!!
DoStatistics	logical	.false.	switch on/off statistical output of init routines
user_numin	real	-99.9	user given value for numin, overrides default value if reasonable
user_numax	real	-99.9	user given value for numax, overrides default value if reasonable
user_costmin	real	-99.9	user given value for costmin, overrides default value if reasonable
user_costmax	real	99.9	user given value for costmax, overrides default value if reasonable
user_ymax	real	-99.9	user given value for ymax, overrides default value if reasonable
user_smin	real	-99.9	user given value for smin, overrides default value if reasonable
user_xBmin	real	-99.9	user given value for xBmin, overrides default value if reasonable
user_qsqmin	real	-99.9	user given value for qsqmin, overrides default value if reasonable
user_qsqmax	real	-99.9	user given value for qsqmax, overrides default value if reasonable
user_maxw	real	-99.9	user given value for maxw, overrides default value if reasonable

earlyPauli	logical	.false.	Flag to indicate, whether we should check Pauli blocking already during generation or only at the end. if .false. (default), events will be generated in a first stage without Pauli blocking. This is then tested afterwards. If the generated event is blocked, it will be redone! Thus Pauli blocking does <i>*not*</i> change the total cross section, only the relative strength will be reshuffled. if .true., then blocked events will be excluded from the Monte Carlo decision and the total cross section will be reduced. NOTES The behaviour, if no event at all is possible, is at the moment a little bit unpredictable ;)
equalWeights_Mode	integer	0	possible values are: • 0: default perweight mode is used (default) • 1: default perweight mode is used, but max is printed • 2: MC rejection method is used. In the default mode, the perweights of the final particles are given by cross section/(A * numEnsembles) If equalWeightsMode==2, then the perweights are given by equalWeights_Max/(A * numEnsembles)
equalWeights_Max	real	-1e99	The maximum value the MC-rejection method is done against.

HiPhotonKinematics code/init/ElectronGenerator/eN_event.f90			
nu	real	-99.9	Photon energy [GeV]
Q2	real	-99.9	transfer four momentum squared [GeV ²]
eps	real	-99.9	Photon polarisation [1]
srts	real	-99.9	sqrt(s) of electron nucleon system [GeV]
W	real	-99.9	sqrt(s) of photon nucleon system [GeV]
xBj	real	-99.9	Bjorken x [1]
Ebeam	real	-99.9	energy of electron beam [GeV]
theta	real	-99.9	scattering angle of electron beam (in lab frame) [°]

HiPion_Analysis code/analysis/HiPionAnalysis.f90			
Enable	logical	.true.	If .true. the HiPion analysis will be performed, otherwise not.
EnablePerTime	logical	.false.	If .true. the HiPion analysis per timestep will be performed, otherwise not.
DoSimpleKin	logical	.false.	switch on/off: Analysis for simple kinematics: pZ-, pT-spectra etc.
DoHarp	logical	.false.	switch on/off: Analysis for the HARP experiment
DoBlobel	logical	.false.	switch on/off: Analysis according Blobel et al.
DoInvMasses	logical	.false.	switch on/off: reporting of pairwise-invariant-masses
DoDOmega	logical	.false.	switch on/off: Analysis for dSigma/dOmega
DoSigmaDetail	logical	.true.	switch on/off: print of sigma_abs, sigma_CX, sigma_DCX
DoOutChannels	logical	.false.	switch on/off: reporting of all final state channels
DoCountParts	logical	.false.	switch on/off: print out multiplicities of particles at every timestep (cf. HeavyIonAnalysis/countParts and HeavyIonAnalysis/useSet)
HiPionNucleus code/init/initHiPion.f90			
distance	real	15.	Distance in z-direction from the nucleus center in fm, where the projectiles are initialized. If negative, the distance will be chosen automatically.
impact_parameter	real	0.	Impact parameter of the projectiles in fm. If positive (or zero), this fixed value is used for all projectiles. If negative, the impact parameter is chosen by Monte Carlo, so that the projectiles are distributed over a certain disk. Cf. 'setPosition'.
ProjectileCharge	integer	0	Charge of projectile particles.
ProjectileID	integer	pion	ID of projectile particles.
ProjectileAnti	logical	.false.	Antiparticle flag of projectile particles.
ProjectileMass	real	-99.9	Mass of projectile particles; if negative, pole mass is used.
nTestparticles	integer	200	Number of projectile testparticles per ensemble.
ekin_lab	real	-99.9	Kinetic energy of projectile particles in lab frame [GeV].
p_lab	real	-99.9	Momentum of projectile particles in lab frame [GeV/c].

DoPerturbativeInit	logical	.false.	If this flag is set to .true., the first collision of the projectile particles with a nucleon in the target nucleus will be done in this init routine (at timestep 0). This enables you to treat the first (hard) collision different from those in the FSI. If this flag is set to .false., the projectile particles have to be propagated onto the nucleus as in the default transport treatment. See documentation of 'initHiPionInducedCollide' and 'initHiPionInducedCollideFull' for further information.
DoOnlyOne	logical	.true.	If the first interaction of beam and target particles is treated already here in the init (cf. DoPerturbativeInit), you may select whether a beam particle may interact only once (flag set to .true.) or with all other target nucleons (flag set to .false.). See documentation of 'initHiPionInducedCollide' and 'initHiPionInducedCollideFull' for further information.
minimumMomentum	real	1.0	Minimal momentum of particles (in GeV) produced in the init routines. Only particles with an absolute momentum larger than this will be further propagated.
useHermesPythiaPars	logical	.false.	flag: Use "PYTHIA tuning done by HERMES collab"
NucCharge	integer	-1	Select charge state of nucleons to scatter on. If this value is ≥ 0 , then we only scatter on nucleons with the respective charge, i.e. only on neutrons if NucCharge==0 and only on protons if NucCharge==1. Useful e.g. for selecting only pn events in a pd collision.
flagOffShell	logical	.false.	If this flag is set to .true., the struck nucleon in the deuteron will be off vacuum mass shell to agree with total energy conservation. Relevant for the deuteron target only.
flagFlux	logical	.false.	If this flag is set to .true., the Moeller flux factor will be included in calculation of the cross section with nuclear target.
flagLC	logical	.false.	If this flag is set to .true., the deuteron wave function will be treated according to the light cone formalism.
CoulombCorrection	logical	.false.	if .true. then a Coulomb propagation from CoulombDistance to distance is performed
CoulombDistance	real	200.	distance from where the Coulomb propagation starts
DoPlotSrts	logical	.false.	flag to switch whether a histogram of the sqrt(s) of the first collision is written or not

History		code/collisions/history.f90	
IncGeneration_Decay	logical	.true.	This flag determines whether we will increase the stored 'generation' of the daughter particles in a resonance decay.
IncGeneration_Elastic	logical	.true.	This flag determines whether we will increase the stored 'generation' of particles in an elastic collision. Setting it to .false. will also prevent elastic collisions from showing up as parents in the history.

InABox		code/init/initInABox.f90	
proton_Density	real	0.084	proton Density [fm ⁻³]
neutron_Density	real	0.084	neutron Density [fm ⁻³]
fermiMotion	logical	.true.	switch on/off Fermi motion
temp	real	0.	If fermiMotion is true, this switch determines the temperature (in GeV) used in the Fermi distribution.
energy_density	real	0.	Energy density in GeV/fm ³ . If a finite positive number is given, the box will be boosted to a frame with the given energy density.
standing_wave_number	integer	0	If this number is larger than zero, the initial density distribution will not be uniform, but is modulated with a standing wave in z direction. The given number determines the number of oscillations throughout the box. The amplitude of the oscillations is currently fixed to be 20% of the (average) nucleon density.

InABoxAnalysis		code/analysis/InABoxAnalysis.f90	
Enable	logical	.true.	Flag to enable or disable the box analysis altogether.
Interval	integer	20	Interval for output, i.e. number of timesteps after which output is written.

InABoxDelta		code/init/initInABoxDelta.f90	
mass	real	1.2	rest mass of Deltas [GeV]
charge	integer	+1	charge of Deltas
mom	real	0.1	momentum of Deltas (in z direction) [GeV]
nDelta	integer	50	number of Deltas per ensemble

initDatabase	code/database/particleProperties.f90		
propagationSwitch	integer	3	• 0 = propagate resonances with more than 1 star in their rating (irrespar=0 in old code) • 1 = propagate just the Delta (irrespar=2 in old code) • 2 = propagate no resonance (irrespar=3 in old code) • 3 = propagate all resonances (default)
usageForXsectionSwitch	integer	2	• 0 = use resonances with more than 1 star rating for cross sections • 1 = use all resonances for cross sections • 2 = use all resonances besides the 1* star I=1/2 resonances • 3 = use only the Delta
rho_dilep	logical	.false.	If .false. (default), the rho meson width will be exclusively given by the 2pi decay and its minmass will be 2m_pi. If .true., the dilepton width will be included in the width and spectral function of the rho, and the minmass will be 2m_e. This is important for dilepton spectra, in order to get contributions from the rho below the 2pi threshold.
FileNameDecayChannels	character(1000)"		The absolute filename of the file containing decay channel infos. possible values: • if not set, default is '[path_To.Input]/DecayChannels.dat' • if given, but does not contain '/': default is '[path_To.Input]/[FileNameDecayChannels]' • otherwise: filename is absolute, including path NOTE if you want to use the file 'XXX.dat' in the actual directory, give it as './XXX.dat'

initDensity	code/density/density.f90		
densitySwitch	integer	1	This switch decides whether the density is static or dynamic during the run. ("Static" makes sense only for fixed target scenarios!) One can use a static density if the nucleus stays roughly in its ground state during the collision. possible values: • 0: Density is set to 0. • 1: Dynamic density according to test-particle distribution. • 2: Static density (not for heavy-ion collisions). • 3: Resting matter: Density is given by the two input parameters "densityInput_neutron" and "densityInput_proton". • 4: Dynamic density in a box. Assumes the same density everywhere, but also calculates the momentum distribution

linearInterpolation	logical	.true.	If this switch is 'true', then the dynamic-density mode uses linear interpolation to determine the density in between the gridpoints.
densityInput_proton	real	0.084	Assumed proton density if densitySwitch=3
densityInput_neutron	real	0.084	Assumed neutron density if densitySwitch=3
gridSize	real, dimension(1:3)	(/12.,12.,12./)	Size of density grid in fm.
gridPoints	integer, dimension(1:3)	(/30,30,30/)	Number of gridpoints in each space direction.
setnewsmearing	logical	.false.	Readjust the smearing to a different width if .true.
newsmearing	real	1.	Use a smearing width as in a grid with newsmearing times the gridspacing
nLargePoints	integer	2	Number of points which are considered to the left and right to smear density on

initNbarN_to_NbarDelta	code/collisions/twoBodyReactions/baryonBaryon/NbarN_to_NbarDelta.f90		
delta_mass	real	0.01	• grid step on a delta mass (GeV)
maxPoints_mass	integer	150	• number of the grid points on the delta mass
delta_srts	real	0.01	• grid step on an invariant energy (GeV)
maxPoints_srts	integer	100	• number of the grid points on the invariant energy

InitNucleus_in_PS	code/init/initNucPhaseSpace.f90		
improvedMC	logical	.false.	• If this flag is set to .true. then we use the information of the already initialized nucleons to decide on the position of a nucleon which has to be initialized. • This prescription only works properly if the smearing width is really small. Therefore it is switched off by default.
improvedMC_speedup	integer	500	• If improvedMC is set to .true. then this variable defines the speedup of the algorithm. • The number defines how often the density field is updated. • A large value of this parameter yields a less accurate test-particle distribution and a faster initialization.
HiTail	logical	.false.	If HiTail is set to .true., then a simple parametrization of $n(p)$ is used to initialize the nucleon momenta (cf. function chooseAbsMomentum for details).

determine_Fermi_momentum_by_binding_energy	logical	.false.	If set to .true., the Fermi momentum will be determined by $E_B = p_F^2 / (2m) + U(\rho, p_F)$, where E_B is the binding energy per nucleon.
determine_Fermi_new_NucDLDA	logical	.false.	If set to .true., the Fermi momentum will be set to a value such that there are no unbound nucleons at the initialisation.
useEnergySF	logical	.false.	If set to .true., then a spectral function is used to choose the energy.
compressedFlag	logical	.false.	If set to .true., then a spherically deformed nucleus is initialized (isotropic compression/expansion; protons and neutrons in phase). This type of deformation corresponds to a giant-monopole resonance mode.
ScaleFactor	real	1.	If compressedFlag=.true., then rescale coordinates by ScaleFactor.
useCdA	logical	.false.	Instead of the usual momentum distribution according a fermi gas, use the momentum parametrizations as given in: • C. Ciofi degli Ati, S. Simula, PRC 53, 1689 (1996) These exist only for 2H,3He,4He,12C,16O,40Ca,56Fe,208Pb
zeroNucleusMomentum	logical	.true.	Indicate whether a procedure should be called to try to find a momentum init where the sum of all nucleon momenta (per ensemble) is zero (or at least close to zero). At the moment, only a hill climbing algorithm is available, which changes the directions of the momenta randomly. The resulting averaged nucleus momentum is in the order of 10 MeV. Without that, the average nucleus momentum goes $\sim 0.17 \text{ GeV} \cdot \sqrt{A}$. (Applies only for $A > 2$.)
printMomDist	logical	.false.	Flag to indicate, whether momentum distribution is written to file

initPauli		code/density/pauliBlocking.f90	
pauliSwitch	integer	1	• 0 : No Pauli blocking • 1 : dynamic Pauli blocking (use actual phase space densities) • 2 : analytic Pauli blocking (use ground state assumption) (not possible for Heavy Ions!) • 3 : dynamic Pauli blocking in a box
densDepMomCutFlag	logical	.false.	if .true. - the radius in momentum space for selecting nucleons around given nucleon will depend on local Fermi momentum NOTES Used only for dynamic pauli blocking.
Gauss	real	1.0	Smearing for dynamic pauli blocking

cutGauss	real	2.2	Cutoff for gauss Smearing
cutMom	real	0.08	• for densDepMomCutFlag=.false. — radius of phase space box in momentum space • for densDepMomCutFlag=.true. — minimum radius of phase space box in momentum space
cutPos	real	1.86	Radius of phase space box in position space
nGridPos	integer	30	number of points in position space to save weights on
ensembleJump	integer	5	Parameter for speedup. Only every "ensemblejump"th ensemble is considered to evaluate the probability for pauli blocking.
DoHistogram	logical	.false.	if .true., a histogram is filled representing the blocking probability as function of the fermi momentum. You have to call 'WriteBlockMom' explicitly for writing the histogram

initRandom code/numerics/random.f90			
Seed	integer	0	Random Seed (used to initialize the random number generator), accessible through the namelist 'initRandom'. If Seed is zero (default), then it is set via "SYSTEM_CLOCK()".
resetRandom	logical	.false.	Reread random generator, used by setRandom, useful for debugging.

initThermoDynamics code/density/thermoDyn.f90			
temperatureSwitch	integer	1	• 1=groundstate calculations (T=0,mu=E_F) • 2=the full procedure
linearExtrapolation	logical	.true.	• .true.= Use linear extrapolation for temperature between gridPoints • .false.= Do not use it

inMedium		code/inputOutput/inputGeneral.f90	
master_piDelta_inmed	logical	.false.	Master switch to turn on or off all the pion-Delta in-med properties, pion potential, 2N pion absorption and Delta inmed broadening NOTES This is a master switch for pi-Delta inmed changes. If set to .true. it turns on all the inmedium pion-Delta corrections, i.e.: • baryonWidthMedium/mediumSwitch=.true., • baryonWidthMedium/mediumSwitch_Delta=.true., • s-wave pion absorption master_3Body/s_wave_abs = .true, and • pion Potential mesonPotential/pionpotSwitch = 4 If .false., the listed switches can be set independently

input		code/inputOutput/inputGeneral.f90	
path_To_Input	character(1000)	'../buuinput'	Path to input files. This switch needs to be set to the local path of the 'buuinput' directory, which contains various input files for GiBUU.
numEnsembles	integer	300	Number of parallel ensembles
eventtype	integer	3	Switch for the type of event possible values: see module eventtypes
fullEnsemble	logical	.false.	Switch for the type of simulation: • .false.=parallel ensembles • .true.=full ensemble See also "localEnsemble".
localEnsemble	logical	.false.	Switch for the type of simulation: • .false. = parallel or full ensembles (depending on the value of the fullEnsemble switch). • .true. = fullEnsemble with "local collisionCriteria", see Lang/Babovsky et al., J. Comput. Phys. 106 (1993) 391-396. Setting localEnsemble = .true. will implicitly set fullEnsemble = .true. (disregarding its value in the jobcard).
delta_T	real	0.2	time difference for time stepping
numTimeSteps	integer	100	Number of time steps
variableTimeStep	logical	.false.	Switch for using of variable time step: • .false.= use constant time step delta_T (see above). • .true.= use time step computed from the frequency of collisions. In this case the input delta_T is used as the maximum allowed time step.

time_max	real	30.	Maximum time until which the time evolution will be computed in the case of variableTimeStep = .true.
num_energies	integer	1	Number of different energies for energy scans
num_runs_sameEnergy	integer	1	Number of runs with the same energy in the initialization.
checkGridSize_Flag	logical	.false.	Switch for checking if particles escape out of grid. possible values: • .false.= no check. • .true. = check is performed, and a warning flag is printed out, in case that particles are outside of the grid. • check valid only for real particles.
continousBoundaries	logical	.false.	• Switch to turn on continous boundary conditions. • Implications for density and propagation. • This means that particles are propagated according to continous boundaries. A particle leaving the grid will move back in from the opposite side. The densities are carefully constructed such that places at the opposite side contribute to places on the near side. • What is still missing is the full implementation in collision criteria, this is not done yet for the two body collisions! Be careful therefore with the 2-Body-collisions at the edges. A particle at one edge does not see its scattering partner at the opposite edge.
FinalCoulombCorrection	logical	.false.	Switch for Coulomb correction at the end of each run of the outgoing particles
length_perturbative	integer	-1	Length of perturbative particle vector (per ensemble). If negative, it will be determined automatically by event type.
length_real	integer	-1	Length of real particle vector (per ensemble). If negative, it will be determined automatically by event type.
freezeRealParticles	logical	.false.	Switch for not propagating real particles
printParticleVectors	logical	.false.	Switch to turn on the printing of the particle vector at the start and end of a run.
printParticleVectorTime	logical	.false.	• Switch to turn on the printing of the particle vector as function of time. • Useful for event classes using real particles (HeavyIon,Hadron). • See also 'timeForOutput' and 'timeSequence'.
printParticleVectorsFormat	integer	1	Select the format for printing the particle vectors. Possible values are: • 1: ASCII • 2: binary
timeForOutput	real	50.	• Time (fm/c) after which the particle vector is printed during run (see also variable "timeSequence"). • valid only if printParticleVectorTime = .true.

timeSequence	real	10.	• Time sequence (fm/c) of time dependent printing of the particle vector • valid only if printParticleVectorTime = .true.
DoPrLevel			
povray_switch	logical	.false.	Switch for generating Povray-Output
LRF_equals_CALC_frame	logical	.false.	• Switch to turn on the assumption that calculation frame and LRF frame coincide • Only useful for reactions close to ground state !!!
DoFragmentNucleons	logical	.false.	• Switch to turn on/off adding of nucleons stemming from fragmentation of bound clusters.
PrintCollisionList	logical	.false.	Switch to turn on the printing of the particles and the collisions continously during the run
onlyFirstEvent	logical	.false.	Switch to turn off all final state colisions by only allowing particles to suffer a 2-body od 3-body collisions, if their field '%firstEvent=0'
version	integer, private	-1	Indicator, for which code version this jobcard is suitable. possible value (at the moment): • 2025: for the use with the release version v2025 if not given, or wrong value, the code will stop execution

input_FF_Delta code/init/lepton/formfactors_Delta/FF_Delta_production.f90			
FF_Delta	integer	1	This switch decides whether the Paschos form factors (FF_Delta=1) or the Maid form factors (FF_Delta=0) are used. Default is FF_Delta=1.

input_FF_ResProd code/init/lepton/formfactors_ResProd/formFactor_ResProd.f90			
FF_ResProd	integer	0	select how the form factors are calculated: • 0: MAID's helicity amplitudes (Luis' helicity expressions - CM frame) • 1: fit of Lalakulich (PRD 74, 014009 (2006)) • 2: MAID's helicity amplitudes (Lalakulich's helicity expressions - LAB frame)
aDelta	real	0	fit parameter for C_5^A (Adler)
bDelta	real	0	fit parameter for C_5^A (Adler)
cDelta	real	3.	fit parameter for C_5^A (Paschos)
C5A0corr	real	0.85	fit parameter for C_5^A (Adler), adjusts strength of C5A0 for the Delta

DeltaAxFF	integer	1	choose between different axial form factors for the Delta: • 1: Adler • 2: Paschos • 3: dipol
HNV_axialFF	logical	.false.	choose axial form factors for the Delta: • .true. is Hernandez-Nieves-Valverde fit with $C_5A=0.867, MA=0.985$ (PRD 76) • .false. is as it was used by Lalakulich et al in PRD 74
nenner_C5A_Lalakulich	real	3.0	Factor which appears in the Lalakulich parameterization of the axial C_5^A form factor: • 3.0 was fitted to BNL and used in Lalakulich PRD71 and PRD 74 • 0.5 is given by a fit of ANL
refit_barnu_axialFF	logical	.false.	if .true., axial form factors are refitted to explain the low value of antineutrino cross section (exp. data Bolognese PLB81,393 (1979))
W_cutOff_lambda	real	1.071	Value for lambda in the W-dependent cut-off function. NOTES cf. https://gibuu.hepforge.org/trac/wiki/Xsections/Compton and J.Weil, PhD thesis, fig. 8
W_cutOff_switch	logical	.false.	Switch to include a W-dependent cut-off function for the vector form factor of the Delta
vector_FF_switch	logical	.true.	Switch to turn off the vector form factors
axial_FF_switch	logical	.true.	Switch to turn off the axial form factors
W_cutOff_switchAll	logical	.false.	Switch to include a W-dependent cut-off function for the vector and the axial form factor of all resonances NOTES For the Delta, the cut off W_cutOff_lambda is used. For all other resonances the pole mass is used, i.e. $\lambda = \text{mass}$
DeltaCouplrelErr	real	0.	error in percent for $C_5^A(0)$ for the Delta
MA	real	1.05	Delta resonance axial mass parameter. Wilkinson et al have shown in Phys.Rev.D 90 (2014) that the ANL pion production data are more reliable than the BNL ones.

insertion code/collisions/insertion.f90			
minimumEnergy	real	0.005	Minimal kinetic energy in GeV for produced perturbative nucleons. If their energy is below this threshold, then they are not propagated, i.e. they are not inserted in the particle vector.
propagateNoPhoton	logical	.true.	If .true. then we eliminate all photons, such that they are not propagated and do not show up in the particle vector. If .false. then photons are explicitly propagated.

Lepton2p2h code/init/neutrino/neutrinoParms.f90				
ME_Version	integer	6	indicate the type of matrix element parametrisation NOTES possible values: • 1: const ME_Norm_XX ! const for CC fitted to MiniBooNE is 1.8e-6 • 2: constant transverse and decreasing with Enu • 3: "Dipole transverse" transverse, fall with Q2 as 4-th power • 4: MEC from E. Christy (8/2015), with parametrization for longitudinal • 5: MEC from Bosted arXiv:1203.2262, with parametrization for longitudinal • 6: MEC additional parametrization, with parametrization for longitudinal not yet implemented remarks: • case 1 is model-I in Lalakulich,Gallmeister,Mosel PRC86(2012)014614 • case 2 is model-II from Lalakulich,Gallmeister,Mosel PRC86(2012)014614 • case 3 gives a good description of MiniBooNE data with MA ~ 1.5 GeV	
ME_Norm_QE	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Overall strength of 2p2h matrix element with 2N out for (EM,CC,NC) NOTES The value == 1 gives the coded strength
ME_Norm_Delta	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Overall strength of 2p2h matrix element with NDelta out for (EM,CC,NC) NOTES The value == 1 is a dummy value
ME_Mass_QE	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Cutoff-mass in some parametrizations of 2p2h matrix element for NN out for (EM,CC,NC) NOTES The value == 1 is a dummy value

ME_Mass_Delta	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Cutoff-mass in some parametrizations of matrix element for NDelta out for (EM,CC,NC) NOTES The value == 1 is a dummy value
ME_Transversity	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Parametrisation of structure functions for (EM,CC,NC) NOTES The value = 1 chooses structure function W2 so that 2p2h is pure transverse
ME_LONG	real, dimension(1:3)	(/0.0, 0.0/)	0.0,	Parametrization of structure functions for (EM,CC,NC) NOTES The value = 0 turns any additional longitudinal contribution to structure funct. W2 off
ME_W3	real, dimension(1:3)	(/0.0, 1.0/)	1.0,	Overall strength factor for structure function W3 only for (CC,NC) NOTES overall strength parameter for structure function W3
inmedW	integer	1		Controls which inv mass W is used in parametrization of 2p2h W1 NOTES • 1: W = static inv. mass in 2p2h parametrization of W1 • 2: W = inv mass for Fermi moving nucleons in potential • 3: W = inv mass for Fermi moving nucleons without potential
Adep	integer	2		Switch for A-dependence of 2p2h structure function NOTES • 1: A-dependence for zero-range force (Mosel, Gallmeister, 2016) • 2: linear A-dependence, normalized to C12

lepton_bin	code/init/neutrino/initNeutrino.f90		
cost_min	real	-1.0	if detailed_diff_output is TRUE: Minimal cos(theta) of outgoing leptons, used in 2D dsigma/dEdcos(theta) This cut affects *only* the outgoing lepton
cost_max	real	+1.0	if detailed_diff_output is TRUE: Maximal cos(theta) of outgoing leptons, used in 2D dsigma/dEdcos(theta) This cut affects *only* the outgoing lepton
delta_cost	real	0.1	if detailed_diff_output is TRUE: stepsize of cos(theta) of outgoing leptons, used in 2D dsigma/dEdcos(theta)
Elept_min	real	0.0	if detailed_diff_output is TRUE: minimal energy of outgoing leptons, used in 2D dsigma/dEdcos(theta)

Elept_max	real	2.0	if detailed_diff_output or printAbsorption are TRUE: maximal energy of outgoing leptons, used in 2D dsigma/dEdcos(theta)
delta_Elept	real	0.01	if detailed_diff_output or printAbsorption are TRUE: stepsize of energy of outgoing leptons, used in 2D dsigma/dEdcos(theta)
pL_min	real	0.0	if detailed_diff or printAbsorption are TRUE: minimal long. momentum of outgoing leptons, used in 2D dsigma/dpLdpT
pL_max	real	20.0	if detailed_diff_output or printAbsorption are TRUE: maximal long. momentum of outgoing leptons, used in 2D dsigma/dpLdpT
delta_pL	real	0.25	if detailed_diff_output or printAbsorption are TRUE: stepsize of long. momentum of outgoing leptons, used in 2D dsigma/dpLdpT
pT_min	real	0.0	if detailed_diff or printAbsorption are TRUE: minimal transv. momentum of outgoing leptons, used in 2D dsigma/dpLdpT
pT_max	real	2.5	if detailed_diff_output or printAbsorption are TRUE: maximal transv. momentum of outgoing leptons, used in 2D dsigma/dpLdpT
delta_pT	real	0.1	if detailed_diff_output or printAbsorption are TRUE: binwidth of transv. momentum of outgoing leptons, used in 2D dsigma/dpLdpT
Q2_Max	real	100.	maximal value of Q2 in Q2-distribution

low_photo_induced code/init/lowPhoton/initLowPhoton.f90			
energy_gamma	real	0.	Energy of incoming photon in nucleus rest frame (in GeV).
delta_energy	real	0.	Increase of energy for energy scans.
energy_weighting	integer	0	Determines the relative weight of different photon energies in energy scans Possible values: • 0 = flat distribution (all energies are weighted equal) • 1 = exponential distr. (energies are weighted $\sim 1/E$)

LowElectron code/init/lowElectron/initLowElectron.f90			
runType	integer	1	• 1: we make runs at some fixed angle defined by initLowElectron/theta_lf. • 2: we make runs at some fixed QSquared defined by initLowElectron/QSquared

inputType	integer	1	Decides which set of variables is used to determine the final electron energy $energy_lf$ and the step size $delta_energy_lf$: • 1: we use directly $energy_lf$ and $delta_energy_lf$ as input • 2: we use W_min and W_max as input. For this we assume the nucleon to be at rest to calculate $energy_lf$ out of W. • 3: we use $energy_lf_min$ and $energy_lf_max$ as input.
theta_lf	real	10.	Theta scattering angle of outgoing electron with respect to the incoming one. Only relevant if $runType=1$.
phi_lf	real	-10.	Phi scattering angle of outgoing electron with respect to the incoming one. If less than 0, then we do a Monte-Carlo-Integration over ϕ !
energy_li	real	1.2	Energy of incoming electron in GeV.
energy_lf	real	0.8	Energy of final state electron in GeV. Only used if $inputType=1$
energy_lf_min	real	0.1	Minimal $energy_lf$ Only used if $inputType=3$
energy_lf_max	real	0.1	Maximal $energy_lf$ Only used if $inputType=3$
delta_energy_lf	real	0.8	$delta(Energy)$ of final state electron in GeV for energy scans. Only used if $inputType=1$
W_min	real	0.9	Minimal W at the hadronic vertex assuming a resting nucleon Only used if $inputType=2$
W_max	real	1.9	Maximal W at the hadronic vertex assuming a resting nucleon Only used if $inputType=2$
QSquared	real	0.5	Q^2 of virtual photon. Only relevant if $runType=2$.
Do_QE	logical	.true.	Switch for including or excluding Quasi-Elastic (QE) processes
Do_1Pi	logical	.true.	Switch for including or excluding direct Single pion production processes. If resonances are included ($Do_Res=.true.$), then only the background part is included.
Do_Res	logical	.true.	Switch for including or excluding resonance production processes
Do_2Pi	logical	.true.	Switch for including or excluding direct Double pion production processes. If resonances are included ($Do_Res=.true.$), then only the background part is included.
Do_DIS	logical	.true.	Switch for including or excluding deeply inelastic scattering (DIS) events Only relevant for $W > 1.4-1.5$ GeV.

Do_2p2hQE	logical	.false.	Switch for including or excluding event according $\gamma^* N_1 N_2 \rightarrow N_1' N_2'$
Do_2p2hDelta	logical	.false.	Switch for including or excluding event according $\gamma^* N_1 N_2 \rightarrow N' \Delta$
minMass_QE	real	0.3	Minimal mass of a nucleon in QE event. Prevents super-luminous nucleons when embedded in a Skyrme potential.
minEnergy_1pi	real	0.16	Minimal q_0 such that pion production processes are considered.
onlyDelta	logical	.false.	Switch for including only delta resonance
nuclearTarget_corr	logical	.true.	• If the input is a nuclear target, then the target nucleus is at rest and we calculate the cross section for nuclear target: use flux with respect to the nucleus. • Use .false. only for debugging.

lowElePhoto_Analysis	code/analysis/lowElectronAnalysis.f90		
dOmega_switch	logical	.false.	If .true. then also dSigma/dOmega is produced, if false not..
dE_switch	logical	.false.	If .true. then also dSigma/dE is produced, if false not..

lowPhotonAnalysis	code/analysis/LowPhotonAnalysis.f90		
outputEvents	logical	.false.	If .true. then all events are printed to file.
outputEvents_onlyFree	logical	.false.	If outputEvents=.true. then only particles which may leave the nucleus, i.e. may become "free", are printed to file.
KruscheOutput	logical	.false.	If .true. then we perform an analysis as in EPJA22 347-351 (2004)
KruscheAnalyse_cut	real	0.	Value of the cut for the deltaE cut in EPJA22 347-351 (2004).
FissumOutput	logical	.false.	If .true. then we perform an analysis as in PRC 53,#3 pages 1278 ff. (1996) Produces dSigma/dOmega/dT_pi for π^+
photonAnalyse	logical	.false.	Special analysis for final state photons
TwoPiOutput	logical	.false.	If .true. then we perform an analysis for 2pi production, including statistics for the mass of the pi-pi pair.
pi0gamma_analysis	logical	.false.	Do analysis of pi0 gamma pairs (dSigma/dm), to reconstruct invariant mass spectrum of omega mesons.
pi0gamma_momcut	real	0.5	Cut on the absolute omega three momentum in GeV, being applied to the pi0 gamma spectrum.

pi0gamma_masscut	real, dimension(1:2)	(/0.,2./)	Cuts on the pi0-gamma invariant mass in GeV, being applied to all pi0-gamma spectra (except the mass spectrum). First component is lower limit, second component is upper limit.
pi0gamma_mombin	real	0.050	Bin size for the pi0 gamma momentum spectrum in GeV.
pi0gamma_massres_sigma	real	0.025	Sigma parameter for the exp. resolution smearing (width of the Gauss or Novosibirsk function in GeV). See also pi0gamma_massres_tau.
pi0gamma_massres_tau	real	-0.090	Skewness parameter tau of the Novosibirsk function (for exp. resolution smearing). See also pi0gamma_massres_sigma.
Ekin_pi0_cut	real	0.	Cut on the kinetic energy of neutral pions in the pi0gamma_analysis. Only pions with kinetic energies larger than this cutoff are used for the analysis.
eta_analysis	logical	.false.	Do analysis of eta mesons
MissingMass_analysis	logical	.false.	Do analysis of missing mass distributions for 1 nucleon and 2 nucleon final states
DoOutChannels	logical	.false.	switch on/off: reporting of all final state channels

MassAssInfo code/typeDefinitions/MassAssInfoDefinition.f90			
UseMassAssInfo	logical	.true.	This switch indicates, whether we want to use the whole MassAssInfo machinery or stick to the old prescription of mass assignment. You may set this switch via the jobcard. Anyhow, if your selection of switches for baryon and medium switches leads to cases which are not yet implemented, this flag is set to false automatically.

master_1body code/collisions/oneBodyReactions/master_1Body.f90			
correctEnergy	logical	.true.	Scale final state momenta to fulfill energy and momentum conservation. If .false. energy conservation is violated
StableInFormation	logical	.true.	Particles during its formation time are considered to be stable or not.
omegaDecayMediumInfo	logical	.false.	Write out information about all decaying omega mesons to a file called "omegaMediumInfo.dat" (decay point, momentum, density, etc).

omegaDecay_restriction	integer	0	This switch, like omegaDecayMediumInfo, helps to analyze $\omega \rightarrow \pi^0 \gamma$ decays. It will only have an effect for omegaDecayMediumInfo = .true. Possible values: • 0 = none (default) • 1 = vacuum ($\rho < 0.1 \rho_0$) • 2 = medium ($\rho > 0.1 \rho_0$) With the default value (0), all omega decays are carried out as usual. For the value 1, the decay products are only kept, if the decay happens in the vacuum (i.e. at $\rho < 0.1 * \rho_0$). For the value 2, the decay products are only kept, if the decay happens in the medium (i.e. at $\rho > 0.1 * \rho_0$). If the density does not meet these conditions, the decay products are simply removed and will not be put in the particle vector (and thus they will not appear in the analysis).
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master_2body	code/collisions/twoBodyReactions/master_2Body.f90		
correctEnergy	logical	.true.	Scale final state momenta to fulfill energy and momentum conservation. If .false. energy conservation is violated
baryonBaryonScattering	logical	.true.	Switch to turn off baryon-baryon-Scattering
baryonMesonScattering	logical	.true.	Switch to turn off baryon-Meson-Scattering
mesonMesonScattering	logical	.true.	Switch to turn off meson-Meson-Scattering
usePythia	integer	1	This flag decides whether to use Fritiof or Pythia for high-energy collisions: • 0: use Fritiof • 1: use Pythia NOTES • This flag is not used in the baryon-antibaryon channel
usePythia_BaB	integer	0	This flag decides whether to use Fritiof or Pythia for high-energy baryon-antibaryon collisions: • 0: use Fritiof • 1: use Pythia
useHiEnergy	logical	.true.	Switch to turn HiEnergy on/off. Formerly known as "useFritiof". NOTES Please be very sure what you are doing when setting this parameter to .false.!
HiEnergyThresholdBarMes	real	2.2	Sqrt(s) threshold for HiEnergy in Baryon-Meson Reactions
HiEnergyThresholdBarMesDelta	real	0.2	width for the Sqrt(s) threshold for HiEnergy in Baryon-Meson Reactions
HiEnergyThresholdBarBar	real	3.4	Sqrt(s) threshold for HiEnergy in Baryon-Baryon Reactions

HiEnergyThresholdBarBarDelta	real	0.1	width for the Sqrt(s) threshold for HiEnergy in Baryon-Baryon Reactions
HiEnergyThresholdBarAntibar	real	2.38	Sqrt(s) threshold for HiEnergy in Baryon-Antibaryon Reactions
HiEnergyThresholdBarAntibarDelta	real	0.0	width for the Sqrt(s) threshold for HiEnergy in Baryon-Antibaryon Reactions
useManni	logical	.true.	Flag, whether to use meson-baryon annihilation as proposed by Markus Wagner (Diploma, Giessen 2004), but with some enhanced treatment
ElastAngDist	integer	3	Choice of angular distribution in (high-energy) elastic collisions (cf. Do-Coll_Elast): • 1 = isotropic • 2 = J. Cugnon et al., NPA 352, 505 (1981) • 3 = Pythia (default)
flagElastBB	logical	.false.	If .true., use a constant elastic baryon-baryon cross section of 40 mb and no inelastic baryon-baryon scattering.
coarse	real, dimension(1:3)	(/3.,4.,4./)	coarse maximal impact parameter (in fm)
bmax_nucleonNucleon	real	2.52	Real maximal impact parameter for nucleon-nucleon-scattering. Maximal crossection is <pre> jpre> bMax**2 * pi * 10 mb/fm**2 = (2.52**2*pi*10) mb = 199.5 mb </pre>
bmax_nucleonResonance	real	1.60	Real maximal impact parameter for nucleon-resonance scattering. Maximal crossection is <pre> jpre> bMax**2 * pi * 10 mb/fm**2 = (1.60**2*pi*10) mb = 80.4 mb </pre>
bmax_hyperonNucleon	real	2.52	Real maximal impact parameter for hyperon-nucleon-scattering. Maximal crossection is <pre> jpre> bMax**2 * pi * 10 mb/fm**2 = (2.52**2*pi*10) mb = 199.5 mb </pre>
bmax_baryonPion	real	2.52	real maximal impact parameter for baryon pion scattering
bmax_baryonMeson	real	2.52	real maximal impact parameter for baryon-Meson-scattering
bmax_mesonMeson	real	2.	real maximal impact parameter for meson-meson-scattering
correctEnergy_message	logical	.true.	Switch off error message for energy correction failures.
OverrideSigma_PiN	real	-99.9	Parameter to replace the calculated cross section for pi+N collision by a fixed value (in mb). Only in use if >= 0. The elastic cross section is assumed to be 1/10 of the given value.
OverrideSigma_RhoN	real	-99.9	Parameter to replace the calculated cross section for rho+N collision by a fixed value (in mb). Only in use if >= 0. The elastic cross section is assumed to be 1/10 of the given value.

OverrideSigma_PiPi	real	-99.9	Parameter to replace the calculated cross section for pi+pi collision by a fixed value (in mb). Only in use if ≥ 0 . We set <code>sigma.elast = sigma.tot</code>
Override_PiPi_ResIsElast	logical	.false.	Flag to replace the calculated cross section for pi+pi collision; The calculated resonant cross section will be transformed into the elastic cross section. Thus no resonances will be propagated explicitly, but they show up in the cross section We set <code>sigma.elast = sigma.Res</code> , <code>sigma.Res = 0</code> , <code>sigma.tot = sigma.elast</code> please note: background processes as $\pi\pi \leftrightarrow K\bar{K}$ are *not* affected by this switch. You have to disable those additionally by hand, see <code>mesMes.do2to2</code>
omega_K_factor	real	2.	Modification factor for the inelastic omega-nucleon cross section. Necessary to describe transparency ratio data measured by CBELSA/TAPS, see: http://arxiv.org/abs/1210.3074
mesMes.do2to2	logical	.true.	flag whether to do $m\bar{m}' \leftrightarrow K\bar{K}, K\bar{K}^*$ etc.
mesMes.useWidth	logical	.false.	flag whether to use the width in $m\bar{m}' \leftrightarrow K\bar{K}, K\bar{K}^*$ etc. This is needed to enforce detailed balance. Otherwise only pole masses are used.
doScaleResidue	logical	.true.	scale the cross section of real-pert collisions by a factor N'/N or Z'/Z for a scattering on a neutron or proton, where N' and Z' are calculated via the residuum.

master_3body <code>code/collisions/threeBodyReactions/master_3Body.f90</code>			
correctEnergy	logical	.true.	Scale final state momenta to fulfill energy and momentum conservation. If .false., energy conservation is violated.
radiusNukSearch	real	2.9	Radius for the search of nucleons, i.e. the radius in which nucleons shall be searched for at <code>rho_0</code> .
DeltaThreeBody	logical	.true.	Switch for the $NN\Delta \rightarrow NNN$ processes (false=OFF).
pionThreeBody	logical	.true.	Switch for the $NN\pi \rightarrow NN$ processes (false=OFF).
positionNNpi	logical	.false.	This switch determines where the final state particles in $NN\pi \rightarrow NN$ are positioned: • true: pion position • false: center of NNpi (default)

s_wave_abs	logical	.false.	Use the s-state absorption of Nieves et al NPA 554 (1993) 554, Eq. (18) for $NN\pi \rightarrow NN$ NOTES • This switch should be .true. when the pion potential pionPot_Oset is used • with pionPot_Switch = 1 or = 4 in nl mesonPotential
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MatrixElementQE code/init/lepton/matrixElementQE.f90			
useQEextraterm	logical	.true.	switch on/off an extra term appearing in the current due to different masses of in- and outgoing nucleons
useCorrelations	logical	.false.	switch on/off RPA correlations according to Nieves, Amaro, Valverde, PRC70, 055503 (2004)
nievesCorr_para	integer	2	if RPA correlations are switched on, this parameter decides which set of variables to use: • 1: modified Nieves et al., PRC70, 055503 (2004) • 2: original Nieves et al., PRC70, 055503 (2004) • 3: Tselyaev, Speth et al., PRC75, 014315 (2007)
gp	real	0.63	vary gp if RPA correlations are switched on
withScalarInt	logical	.true.	switch on/off scalar interactions

MediumModule code/density/medium.f90			
mediumCutOff	real	1.E-8	If the density is lower than this value, then we treat the medium like vacuum.

mesonPotential code/potential/mesonPotential.f90			
pionPot_Switch	integer	0	Switch for pion potential: • 0 = no potential • 1 = Oset potential (NPA 554), which is valid up to 70 MeV kinetic energy • 2 = Kapusta suggestion for pion potential (rather unusual) • 3 = Delta-Hole potential, which is valid up to 130 MeV kinetic energy • 4 = Smooth spline transition between switch 1 and 3. NOTES Can be set in namelist mesonPotential.

noPerturbativePotential	logical	.false.	Switch for potential of perturbative particles. If .true. then perturbative mesons feel no potential. NOTES Can be set in namelist mesonPotential.
vectorMesonPot	integer	0	Switch for medium-modification of vector mesons: • 0 = no modification • 1 = Brown-Rho-Scaling • 2 = Brown-Rho-Scaling with momentum dependence according to Kondratyuk (see page 162 in Effenberger's thesis). Currently not available! NOTES Can be set in namelist mesonPotential.
brownRho	real	0.16	Brown-Rho scaling parameter alpha.

MesonWidthVacuum code/width/mesonWidthVacuum.f90			
omega_width	integer	1	Select a parametrization for the omega vacuum width: • 1 = GiBUU default (a la Manley) • 2 = Muehlich
srts_srt_switch	logical	.false.	Modifies the width according to S. Leupold's definition of the width, one especially has to exchange s against sqrt(s) in the denominator of Formula 2.76 of Effenbergers Phd NOTES The default value is .false. and the power of the mass resp. sqrt(s) is 1. If the flag is .true., the power is 2.

ModifyParticles code/database/particleProperties.f90			
mass	real, dimension(1:pion+nMes-1)	-1.0	Input array for modifications on the particle mass NOTES This array is intended to "input" values for the mass of the particles, which are different from the default. Therefore only entries, which are positive after reading the file are stored in the internal database.
width	real, dimension(1:pion+nMes-1)	-1.0	Input array for modifications on the particle width NOTES This array is intended to "input" values for the width of the particles, which are different from the default. Therefore only entries, which are positive after reading the file are stored in the internal database.

stabilityFlag	integer, -1 dimension(1:pion+nMes-1)	Input array for modifications on the particle stability NOTES This array is intended to "input" values for the stability of the particles, which are different from the default. Therefore only entries, which are >-1 after reading the file are stored in the internal database. The index of the array is the particle ID. The value encodes on a bitwise level, how the particle may decay (cf. also master_1Body): • 1: particle may decay during run, if $\Gamma > \text{gammaCutOff}$ • 2: particle may decay at the end of the run, if $\Gamma > 0$. • 4: particle may decay at the end via Jetset, if there the parameters allow for a decay. The default values are one of the following: • 0: particle may not decay at all (i.e. it is stable) • 3: particle may decay both during run and at the end (combination of 1 and 2)
propagated	integer, -1 dimension(1:pion+nMes-1)	Input array for modifications on the flag propagated NOTES This array is intended to "input" values for the flag of the particles, which are different from the default. Therefore only entries, which are zero or positive after reading the file are stored in the internal database. Here 0 is understood as .false., while all positive values stand for .true.
neutrino_induced code/init/neutrino/initNeutrino.f90		
process_ID	integer 2	Determine the process (cf. module leptonicID): • 1 = EM • 2 = CC • 3 = NC • -1 = antiEM • -2 = antiCC • -3 = antiNC
flavor_ID	integer 2	Determine the lepton flavor: • 1 = electron • 2 = muon • 3 = tau

nuXsectionMode	integer	0	To choose which kind of Xsection is calculated. All values set in module neutrino_IDTable.f90 possible values: • 0 = integratedSigma: required input: enu • 1 = dSigmadCosThetadElepton: required input: enu, costheta, elepton • 2 = dSigmadQ2dElepton: required input: enu, Q2, elepton • 4 = dSigmadCosTheta: required input: enu, costheta • 5 = dSigmadElepton: required input: enu, elepton • 6 = dSigmaMC: required input: enu • 7 = dSigmaMC_dW: required input: enu, W • 3 = dSigmaMC_dQ2: required input: enu, Q2 • 8 = fixedGamma: required input: 3 variables (only EM) calculation for specific experiments taking into account the flux (choose your favorite experiment with flag nuExp): • 10 = EXP_dSigmadEnu • 11 = EXP_dSigmadCosThetadElepton • 12 = EXP_dSigmadQ2dElepton • 14 = EXP_dSigmadCosTheta • 15 = EXP_dSigmadElepton • 16 = EXP_dSigmaMC • 17 = EXP_dSigmaMC_dW • 13 = EXP_dSigmaMC_dQ2
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nuExp	integer	0	• 0 = no specific experiment • 1 = MiniBooNE neutrino flux (in neutrino mode = positive polarity) • 2 = ANL • 3 = K2K • 4 = BNL • 5 = MiniBooNE antineutrino flux (in antineutrino mode = negative polarity) • 6 = MINOS muon-neutrino in neutrino mode • 7 = MINOS muon-antineutrino in neutrino mode • 8 = NOVA neutrino (medium energy NuMI, 14 mrad off-axis), FD • 9 = T2K neutrino off-axis 2.5 degrees (at ND280 detector) • 10 = uniform distribution from Eflux_min to Eflux_max (see namelist nl_neutrino_energyFlux in the module expNeutrinoFluxes) • 11 = MINOS muon-neutrino in antineutrino mode • 12 = MINOS muon-antineutrino in antineutrino mode • 13 = MINERvA muon neutrino, old flux • 14 = MINERvA muon antineutrino, old flux • 15 = LBNF/DUNE in neutrino mode • 16 = LBNF/DUNE in antineutrino mode • 17 = LBNO neutrino in neutrino mode • 18 = NOMAD • 19 = BNB nue BNB= Booster Neutrino Beam • 20 = BNB nuebar • 21 = BNB numu • 22 = BNB numubar • 23 = NOvA ND • 24 = T2K on axis • 25 = MINERvA, 2016 flux • 26 = FASERnu • 99 = user provided input file
includeQE	logical	.true.	include QE scattering
includeDelta	logical	.true.	include Delta excitation
includeRes	logical	.true.	include excitation of higher resonances
include1pi	logical	.false.	include one-pion cross section see neutrinoXsection.f90 for details: there one might choose between different models and also whether it is taken as background or as total cross section
include2pi	logical	.false.	include 2 pion background channel

includeDIS	logical	.false.	include DIS contribution
include2p2hQE	logical	.false.	include 2p2h QE contribution
include2p2hDelta	logical	.false.	include 2p2h Delta contribution
sigmacut	real	10e-4	events with a cross section smaller than this value are skipped.
realRun	logical	.false.	Do not initialize the final state particles as perturbative particles but as real ones.
printAbsorptionXS	logical	.false.	flag to produce output about inclusive (absorption) cross sections
printInclHist	logical	.true.	flag to produce additional output about inclusive cross sections only checked, if printAbsorptionXS = T
FileNameFlux	character(1000)"		The absolute filename of the file containing flux info, if user supplied possible values: • if given, but does not contain '/': default is '[path.To.Input]/[FileNameFlux]' • otherwise: filename is absolute, including path ('~' is okay) NOTE if you want to use the file 'XXX.dat' in the actual directory, give it as './XXX.dat'
Enumax	real	100.	maximum of neutrino energy in flux distribution, in GeV
delta.Enumax	real	0.02	bin width of neutrino energy in flux distribution, in GeV
storeNucleon	integer	2	indicate which kind of struck nucleon to save: • 1: free Nucleon (i.e. potential removed) • 2: bound nucleon NOTES real check of energy and momentum conservation only possible with '2'
equalWeights_Mode	integer	0	possible values are: • 0: default perweight mode is used (default) • 1: default perweight mode is used, but max is printed • 2: MC rejection method is used. In the default mode, the perweights of the final particles are given by cross section/(A * numEnsembles) If equalWeightsMode==2, then the perweights are given by equalWeights_Max/(A * numEnsembles) Please check in the output the line "numberOfSuccess = ..." for the number of events actually generated.
equalWeights_Max	real	-1e99	The maximum value the MC-rejection method is done against.

neutrino_MAIDlikeBG	code/init/neutrino/singlePionProductionMAIDlike.f90		
b_proton_pinull	real	3.	strength of 1pi BG for CC, multiplies EM BG 3. is tuned to ANL, 6. is tuned to BNL
b_neutron_piplus	real	1.5	strength of 1pi BG for CC, multiplies EM BG 1.5 is tuned to ANL, 3. is tuned to BNL

neutrino_matrixelement	code/init/neutrino/NeutrinoMatrixElement.f90		
which_resonanceModel	integer	0	to change between different realizations of the matrix elements: • 0 = with Fortran calculated matrix elements containing all resonances (default) • 1 = with Mathematica calculated matrix elements (only Delta), no longer maintained • 2 = Rein and Sehgal's matrix elements

NeutrinoAnalysis	code/analysis/neutrinoAnalysis.f90		
detailed_diff_output	logical	.false.	If .true. then also the detailed output of differential cross sections is produced
include_W_dist	logical	.false.	If .true. then the invariant mass distributions for events with 1 pion and 1 nucleon in the final state are produced
kineticEnergyDetectionThreshold_lepton	real	0.0	kineticEnergyDetectionThreshold only lepton kinetic energies above this threshold can be detected This cut affects *all* events, not just the outgoing lepton!
AngleUpperDetectionThresholdDegrees_lepton	real	180.0	lepton angles up to this value can be detected This cut affects *all* events, not just the outgoing lepton!
kineticEnergyDetectionThreshold_nucleon	real	0.0	kineticEnergyDetectionThreshold lower detection threshold for nucleon kinetic energies
AngleUpperDetectionThresholdDegrees_nucleon	real	180.0	nucleon angles up to this value can be detected
kineticEnergyDetectionThreshold_chargedpion	real	0.0	kineticEnergyDetectionThreshold
AngleUpperDetectionThresholdDegrees_chargedpion	real	180.0	charged pion angles up to this value can be detected
kineticEnergyDetectionThreshold_neutralpion	real	0.0	kineticEnergyDetectionThreshold
AngleUpperDetectionThresholdDegrees_neutralpion	real	180.0	neutral pion angles up to this value can be detected

applyCuts	integer	0	This parameter encodes 'binary', which cuts should be applied: • 1: lepton_acceptance • 2: isBound • 4: isBelowThreshold Instead of having three independent flags (with values=0 or 1) as e.g. labelled 'doLepton', 'doIsBound', 'doBelowThr', applyCuts combines them formally into one number as <pre>ipre_i applyCuts = 1*doLepton + 2*doIsBound + 4*doBelowThr</pre> So by setting any number between 0 and 7, one can individually switch on and off each of these cuts. 'lepton_acceptance' uses the input parameters: • kineticEnergyDetectionThreshold_lepton (for all kind of outgoing leptons) • AngleUpperDetectionThresholdDegrees_lepton 'isBound' tests, whether kinetic energy plus potential is <0 'isBelowThreshold' uses the input parameters: • kineticEnergyDetectionThreshold_lepton (only for muons) • AngleUpperDetectionThresholdDegrees_lepton (only for muons) • kineticEnergyDetectionThreshold_nucleon (only for nucleons) • AngleUpperDetectionThresholdDegrees_nucleon (only for nucleons) • kineticEnergyDetectionThreshold_chargedpion (only for charged pion) • AngleUpperDetectionThresholdDegrees_chargedpion (only for charged pion) • kineticEnergyDetectionThreshold_neutralpion (only for neutral pions) • AngleUpperDetectionThresholdDegrees_neutralpion (only for neutral pions) Some examples: • To generate full inclusive output, set the value applyCuts=0 • To generate output where bound nucleons are dropped, set applyCuts=2 • To generate output with specific experimental cuts for the outgoing hadrons, set applyCuts=4 or applyCuts=6 and set the corresponding threshold parameters accordingly. • If in the experiment also cuts on the outgoing lepton are used, set applyCuts=7 and set the corresponding threshold parameters accordingly. NOTES These cuts affects the output into the file "FinalEvents.dat". The cut 'lepton_acceptance' (de-)selects the full event, while the other two cuts only decide whether a specific particle is accepted or not. The kinetic energy of a bound nucleon is < 0. Therefore using the default value kineticEnergyDetectionThreshold_nucleon=0.0 also tests, whether the particle is bound or not. Set the parameter to a large negative value to become ineffective.
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Fissum_analysis	logical	.false.	do analysis with cuts as needed for Fig 25 in Fissum et al, PRC 70, 034606 (2004)
ZeroPion_analysis	logical	.false.	produce output of xsec for various final states with 0 pions and 2 pions see file see sigma_0pions.dat for the list of the final states see files neutrino_0pions.dat, neutrino_0pions.QE.dat, neutrino_0pions.Delta.dat, ... for output
calorimetric_analysis	logical	.false.	do calorimetric energy-transfer and neutrino-energy reconstruction (for each QE, Delta, ...)
radialScale	real	1.5	If radial position of nucleon < radialScale*target radius, then the nucleon is assumed to be bound
reconstruct_neutrino_energy	logical	.false.	reconstruct neutrino energy for final state in "specificEvent_analysis" NOTES .true. must be combined with specificEvent_analysis=.true. and at least one specific event .true.
outputEvents	logical	.false.	If .true. then all events are written to the file 'FinalEvents.dat'.
outputEvents_lepIn	logical	.false.	If outputEvents == .true., then also the incoming lepton is written as a documentation line if this flag is .true.
outputEvents_Residuum	logical	.false.	If outputEvents == .true., then also the residuum is written as a documentation line if this flag is .true.
specificEvent_analysis	logical	.false.	do analysis for specific final states
Xsection_analysis	logical	.false.	If .true. then files "...total_Xsection..." and "...dSigmadEkin..." are printed.
doPipe	logical	.false.	If .true. then events are not written to the file 'FinalEvents.dat', but insted written into a named pipe (fifo) with the name fileNamePipe.
fileNamePipe	character(len=1000)		name of the pipe to be used
DoOutChannels	logical	.false.	switch on/off: reporting of all final state channels
OutChannelsFak	real	1.0	additional scaling of the cross section output of OutChanbels analysis

nl_calorimetric_analysis code/analysis/neutrinoAnalysis.f90			
numin	real	0.	for calorimetric analysis: values for transferred energy; only work if calorimetric_analysis is .true. set the min, max and bins for nu distributions
numax	real	10.0	for calorimetric analysis: values for transferred energy; only work if calorimetric_analysis is .true. set the min, max and bins for nu distributions

nubin	real	0.1	for calorimetric analysis: values for transferred energy; only work if calorimetric_analysis is .true. set the min, max and bins for nu distributions
Enumin	real	0.	for calorimetric analysis: values for neutrino energy; only work if calorimetric_analysis is .true. set the min, max and bins for nu distributions
Enumax	real	10.0	for calorimetric analysis: values for neutrino energy; only work if calorimetric_analysis is .true. set the min, max and bins for nu distributions
Enubin	real	0.1	for calorimetric analysis: values for neutrino energy; only work if calorimetric_analysis is .true. set the min, max and bins for nu distributions

nl_dSigmadcostheta code/init/neutrino/neutrinoSigma.f90			
enu	real	-10.	neutrino energy, read in by namelist
costheta	real	-10.	cosine of the angle between the neutrino (z-direction) and the outgoing lepton
delta_costheta	real	-10.	value by which costheta is increased
integralPrecision	integer	3	precision for the Gauss integration (reduce it for nuXsectionMode.eq.0 (sigma) to e.g. 2)
integralPrecisionQE	integer	500	precision for the Gauss integration over the QE peak (reduce it for nuXsection-Mode.eq.0 (sigma) to e.g. 300)

nl_dSigmadCosThetadElepton code/init/neutrino/neutrinoSigma.f90			
enu	real	-10.	neutrino energy, read in by namelist
costheta	real	-10.	cosine of the angle between the neutrino (z-direction) and the outgoing lepton
elepton	real	-10.	energy of the outgoing lepton
delta_elepton	real	-10.	value by which elepton is increased

nl_dSigmadElepton code/init/neutrino/neutrinoSigma.f90			
enu	real	-10.	neutrino energy, read in by namelist
elepton	real	-10.	energy of the outgoing lepton
delta_elepton	real	-10.	value by which elepton is increased

integralPrecision	integer	3	precision for the Gauss integration (reduce it for nuXsectionMode.eq.0 (sigma) to e.g. 2)
integralPrecisionQE	integer	500	precision for the Gauss integration over the QE peak (reduce it for nuXsection-Mode.eq.0 (sigma) to e.g. 300)

nl_dSigmadQ2	code/init/neutrino/neutrinoSigma.f90		
enu	real	-10.	neutrino energy, read in by namelist
Q2	real	-10.	momentum transfer squared
delta_Q2	real	-10.	value by which Q2 is increased

nl_dSigmadQ2dElepton	code/init/neutrino/neutrinoSigma.f90		
enu	real	-10.	neutrino energy, read in by namelist
Q2	real	-10.	momentum transfer squared
elepton	real	-10.	energy of the outgoing lepton
delta_elepton	real	-10.	value by which elepton is increased

nl_dSigmadW	code/init/neutrino/neutrinoSigma.f90		
enu	real	-10.	neutrino energy, read in by namelist
W	real	-10.	invariant mass defined as $(p+q)^2$
delta_W	real	-10.	value by which W is increased

nl_fluxcuts	code/init/neutrino/esample.f90		
Enu_lower_cut	real	0.	cut events with neutrino energy below Enu_lower_cut; for ANL experiment, for example, Enu_lower_cut=0.5 for some analysis of ppi+
Enu_upper_cut	real	10000.	cut events with neutrino energy above Enu_upper_cut; for ANL experiment, for example, Enu_upper_cut=1.5 for ppi0 and npi+ final state, but 5.98 for ppi+, 6000 for FASER

energylimit_for_Qsrec	logical	.false.	switch for using the energylimits Enu_upper_cut and Enu_lower_cut for the Q ² reconstruction; values: .true. or .false. (default: .false.)
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nl_integratedSigma	code/init/neutrino/neutrinoSigma.f90		
enu	real	-10.	neutrino energy, read in by namelist
delta_enu	real	-10.	value by which the neutrino energy is increased
integralPrecision	integer	3	precision for the Gauss integration (reduce it for nuXsectionMode.eq.0 (sigma) to e.g. 2)
integralPrecisionQE	integer	500	precision for the Gauss integration over the QE peak (reduce it for nuXsection-Mode.eq.0 (sigma) to e.g. 300)

nl_Neutrino2piBack	code/init/neutrino/neutrinoParms.f90		
NormpiBG	real	1.0	overall normalization factor for pi BG and Bloom-Gilman X-section, only relevant for neutrinos
normRES			
normBC	real	1.0	overall normalization factor for neutrino-induced Christy-Bosted contributions between 2 GeV and DIS onset only relevant for neutrinos

nl_neutrino_energyFlux	code/init/neutrino/expNeutrinofluxes.f90		
Eb	real	0.034	constant binding energy used for energy and Q ² reconstruction based on QE scattering kinematics
Eflux_min	real	0.2	minimum energy for uniform flux distribution minimum and maximum energies for the uniform neutrino flux (nu-Exp=10 in the namelist neutrino_induced) can be changed in the namelist nl_neutrino_energyFlux
Eflux_max	real	2.5	maximum energy for uniform flux distribution minimum and maximum energies for the uniform neutrino flux (nu-Exp=10 in the namelist neutrino_induced) can be changed in the namelist nl_neutrino_energyFlux

nl_neutrinoxsection	code/init/neutrino/neutrinoXsection.f90		
singlePiModel	integer	1	to change between different models for the pion nucleon cross section: • 0 = pi N according to Nieves et al (hep-ph/0701149) • 1 = MAID-like model • 2 = Bosted-Christy
invariantMassCut	real	100.	cut events with invariant Mass above this value (in GeV); cut pion production from Delta and DIS on $W_{rec} = \sqrt{M^2 + 2M^*nu - Q^2}$
invariantMassCut_BG	real	100.	cut MAID-like background events with invariantMass_BG above this value (in GeV); cut 1pi BG on $W_{rec} = \sqrt{M^2 + 2M^*nu - Q^2}$
DIScutW	real	2.9	W-cut for sigmoid onset of DIS
DIScutwidth	real	0.05	width for sigmoid onset of DIS
REScutW	real	2.0	W-cut for end of 1pi,2pi BGs
Q2cutoff	logical	.true.	cuts low Q^2 for $W > 3$ GeV
DISformfakEM	integer	2	Introduce an additional form factor for the DIS cross section, when processed via a photon: • 0: no form factor • 1: $Q^2/(mcutDIS^2+Q^2)$ • 2: $Q^4/(mcutDIS^2+Q^2)^2$ In case of electron induced events, we need choose 2 in order to be compatible with Pythia's electron machinery.
DISformfakNCCC	integer	1	Introduce an additional form factor for the DIS cross section, when processed via W or Z boson: • 0: no form factor • 1: $Q^2/(mcutDIS^2+Q^2)$ • 2: $Q^4/(mcutDIS^2+Q^2)^2$ In case of electron induced events, we need choose 2 in order to be compatible with Pythia's electron machinery.
mcutDIS	real	0.34	parameter to control Q^2 dependence of DIS
DISrespectHad	logical	.true.	Flag to indicate, whether hadronization failures should be respected and affect the overall DIS cross section Pythia is run to generate the DIS cross section. But not every of the generated events may lead to a correct hadronic final state.
DISdoMSTP23	logical	.true.	Flag to indicate, whether in Pythia for neutrino-DIS the value MSTP(23)=1 should be used or not
new2piBG	logical	.true.	Flag to turn on the new treatment of 2pi BG for electrons and neutrinos

indBG	integer	3	Index to choose Bloom-Gilman like BG parametrization 1 : original Bloom Gilman 2 : Niculescu fit 3 : nonresonant BG fit from Christy-Bosted
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nl_neweN	code/init/neutrino/neutrinoParms.f90			
ME_Version	integer	6		indicate the type of matrix element parametrisation NOTES possible values: • 1: const ME_Norm_XX ! const for CC fitted to MiniBooNE is 1.8e-6 • 2: constant transverse and decreasing with Enu • 3: "Dipole transverse" transverse, fall with Q2 as 4-th power • 4: MEC from E. Christy (8/2015), with parametrization for longitudinal • 5: MEC from Bosted arXiv:1203.2262, with parametrization for longitudinal • 6: MEC additional parametrization, with parametrization for longitudinal not yet implemented remarks: • case 1 is model-I in Lalakulich,Gallmeister,Mosel PRC86(2012)014614 • case 2 is model-II from Lalakulich,Gallmeister,Mosel PRC86(2012)014614 • case 3 gives a good description of MiniBooNE data with MA $\sim$ 1.5 GeV
ME_Norm_QE	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Overall strength of 2p2h matrix element with 2N out for (EM,CC,NC) NOTES The value == 1 gives the coded strength
ME_Norm_Delta	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Overall strength of 2p2h matrix element with NDelta out for (EM,CC,NC) NOTES The value == 1 is a dummy value
ME_Mass_QE	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Cutoff-mass in some parametrizations of 2p2h matrix element for NN out for (EM,CC,NC) NOTES The value == 1 is a dummy value
ME_Mass_Delta	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Cutoff-mass in some parametrizations of matrix element for NDelta out for (EM,CC,NC) NOTES The value == 1 is a dummy value

ME_Transversity	real, dimension(1:3)	(/1.0, 1.0/)	1.0,	Parametrisation of structure functions for (EM,CC,NC) NOTES The value = 1 chooses structure function W2 so that 2p2h is pure transverse
ME_LONG	real, dimension(1:3)	(/0.0, 0.0/)	0.0,	Parametrization of structure functions for (EM,CC,NC) NOTES The value = 0 turns any additional longitudinal contribution to structure funct. W2 off
ME_W3	real, dimension(1:3)	(/0.0, 1.0/)	1.0,	Overall strength factor for structure function W3 only for (CC,NC) NOTES overall strength parameter for structure function W3
inmedW	integer	1		Controls which inv mass W is used in parametrization of 2p2h W1 NOTES • 1: W = static inv. mass in 2p2h parametrization of W1 • 2: W = inv mass for Fermi moving nucleons in potential • 3: W = inv mass for Fermi moving nucleons without potential
Adep	integer	2		Switch for A-dependence of 2p2h structure function NOTES • 1: A-dependence for zero-range force (Mosel, Gallmeister, 2016) • 2: linear A-dependence, normalized to C12

nl_NievesHadronTensor		
	code/init/neutrino/NievesHadronTensor.f90	
DeltaPole	logical	.true.
crossedDelta	logical	.true.
nucleonPole	logical	.true.
crossedNucleonPole	logical	.true.
contactTerm	logical	.true.
pionPole	logical	.true.
pionInFlight	logical	.true.

nl_SigmaMC	code/init/neutrino/neutrinoSigma.f90		
enu	real	-10.	neutrino energy, read in by namelist
MC_xmax	real	2.0	

nl_singlePionProductionNHVlike	code/init/neutrino/singlePionProductionNHVlike.f90		
integrate_over	integer	2	possible values: • 1 = costhetaPi • 2 = Epi • 3 = over cosThetaPi_star.qz in CM frame which 3-pl differential cross sectio to use for integration: • 1= dsigma/dcostheta/dElepton/dcosThetaPion was originally used and works for nuclei. disadvantage: for some cosThetaPion there are two solutions for Epi, this leads to fluctuations on the cross section • 2= dsigma/dcostheta/dElepton/dEPion has an advantage, that for a given pion energy there is only one solution for the angle between the resonance and pion momenta. so the integration is simpler and results should be smoother NOTES • for 1 : the only option checked for nucleus • for 2 : code works better and faster, gives significantly smoother results below Delta peak. disadvantage: now for the free nucleon only, TO DO : nuclei

nl_specificEvent	code/analysis/neutrinoAnalysis.f90		
no_pi	logical	.false.	do analysis for specific final states: specificEvent=1, no_pi (for example, for QE-like MiniBooNE)
p_Xn_no_pi	logical	.false.	do analysis for specific final states: specificEvent=2
piplus	logical	.false.	do analysis for specific final states: specificEvent=3, 1 pi+ X nucleons mesons of other flavor
piplus_MULTI	logical	.false.	do analysis for specific final states: specificEvent=4 >=1 pi+ X other pions (incl pi+) X nucleons
pi0	logical	.false.	do analysis for specific final states: specificEvent=5, 1 pi0 X nucleons, plus mesons of other flavor
pi0_MULTI	logical	.false.	do analysis for specific final states: specificEvent=6, >=1 pi0 X other pions X nucleons, (pi0 K2K)

piminus	logical	.false.	do analysis for specific final states: specificEvent=7 1 pi- X other pions X nucleons
piminus_MULTI	logical	.false.	do analysis for specific final states: specificEvent=8 >=1 pi- X other pions X nucleons
pp_no_pi	logical	.false.	do analysis for specific final states: specificEvent=9 2 protons, X neutrons, 0 pions
pn_no_pi	logical	.false.	do analysis for specific final states: specificEvent=10 1 neutron, 1 proton, 0 pions
nn_no_pi	logical	.false.	do analysis for specific final states: specificEvent=11 2 neutrons, X protons, 0 pions
pp_Xn_no_pi	logical	.false.	do analysis for specific final states: specificEvent=12 2 protons, X neutrons, 0 pions
nn_Xp_no_pi	logical	.false.	do analysis for specific final states: specificEvent=13 2 neutrons, X protons, 0 pions
ppp_Xn_no_pi	logical	.false.	do analysis for specific final states: specificEvent=14 3 protons, X neutrons, 0 pions
pppp_Xn_no_pi	logical	.false.	do analysis for specific final states: specificEvent=15 4 protons, X neutrons, 0 pions
p_no_pi	logical	.false.	do analysis for specific final states: specificEvent=16 1 proton, 0 neutron, 0 pion
n_no_pi	logical	.false.	do analysis for specific final states: specificEvent=17 1 neutron, 0 proton, 0 pion
Xn_no_pi	logical	.false.	do analysis for specific final states: specificEvent=18, 0 proton, X neutrons, 0 pions
excl_pi0			
excl_piplus			
excl_piminus			
full_incl	logical	.true.	do analysis for specific final states: specificEvent=22 fully inclusive event, all hadrons in final state
binsizeQ2	real	0.01	do analysis for specific final states: binning for reconstruction of Q2 and Enu
binsizeEnu	real	0.02	do analysis for specific final states: binning for reconstruction of Q2 and Enu
maxQ2	real	5.0	do analysis for specific final states: max values for reconstruction of Q2 and Enu
maxEnu	real	5.0	do analysis for specific final states: max values for reconstruction of Q2 and Enu

excl_hadron	logical	.false.	do analysis for specific final states: specificEvent=19,20,21 exclusive 1 pion, no other pions or other mesons of different flavor There could be still other mesons which are heavier than the D, Such events (very rare at DUNE energies) could be counted as exclusive single-meson cross section. This could be cured by extending the list of stable mesons
QEp	logical	.false.	if .true, do analysis for specific analysis for QE-like event with 1 mu, 0 pi, X p

OffShellPotential code/width/offShellPotential.f90			
useOffShellPotentialBaryons	logical	.false.	Switch on or off whether the offshellness should be used for baryons. NOTES • must be set to "TRUE" if mediumSwitch_coll (see module Baryon-WidthMedium) is .true. • if .true. then delta_T (see module inputGeneral) must be ≤ 0.05 AND delta_P (see module propagation) must be ≤ 0.002; AND delta_E (see module propagation) must be ≤ 0.002; slows down propagation by a factor of 10
useOffShellPotentialMesons	logical	.false.	Switch on or off whether the offshellness should be used for mesons.
extrapolateBaryonWidth	logical	.true.	Whether to extrapolate the baryon width below minimal mass or not.
max_offshellparameter	real	5.	The maximal value for the offshell parameter. Note: empirical value! This only applies to baryons. For mesons we have no restrictions on the offshell parameter.
relativistic	logical	.false.	• false: Use non-rel. off-shell parameter $x = \Delta m / \Gamma$, which obeys Stefan Leupold's non-rel. EOM. • true: Use rel. off-shell parameter $x = \Delta m^2 / \Gamma$, which obeys Cassing's rel. EOM.
SetOffShellEnergyFlag	logical	.false.	• false: the energy of off-shell particle is constant during time evolution (static nucleus) • true: the energy of off-shell particle varies during time evolution (dynamic case, e.g. heavy ion collision)

paramEP code/init/lowElectron/ParamEP.f90			
useParam	integer	2	select, which parametrization to use when CalcParamEP is called: • 1: Brasse • 2: Bosted

photonXS code/collisions/twoBodyReactions/HiEnergy/photonXS.f90			
iParam	integer	2	Switch to select the kind of parametrization for $\gamma N \rightarrow V N$: • 1: "old parametrization", fit to experimental data, cf. Effenberger PhD, p.53 • 2: Pythia, cf. Friberg/Sjöstrand hep-ph/0007314 • 3: Donnachie, Landshoff [citation needed]
omega_saphir	logical	.true.	If .true. an improved fit (to SAPHIR data) will be used for $\gamma N \rightarrow \omega N$. cf. "calcXS_omega_saphir"

pionAnalysis code/analysis/LoPionAnalysis.f90			
CMFrame	logical	.false.	If .true. Xsection is evaluated in CM-Frame of the incoming pion and a resting nucleon, else in calculation frame.
dsigma_dOmegadE_switch	logical	.false.	If .true. then dsigma/dOmega and dSigma/dOmega/dE are evaluated.
twoPi_switch	logical	.false.	If .true. then 2Pi output is evaluated.

pionNucleus code/init/initPion.f90			
UseCoulomb	logical	.false.	if .true. then a Coulomb propagation from CoulombDistance to distance is performed
CoulombDistance	real	200.	distance from where the Coulomb propagation starts
distance	real	15.	initialization distance
impact_parameter	real	0.	impact parameter. If less than 0, than an impact parameter integration is performed
charge	integer	0	charge of pion
numberPions	integer	200	number of initialized pions per ensemble
ekin_lab	real	0.	kinetic energies of pions in lab frame.
delta_ekin_lab	real	0.01	step size for kinetic energies in energy scans

pn_medium code/width/proton_neutron_width_medium.f90			
density_dependent	logical	.false.	the density of the spectral function
pn_medium_switch	logical	.true.	If .true. medium_modifications will be used

form_factor	logical	.true.	If .true. the form factor for the width is used
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projectile	code/density/nucleus.f90		
A	integer	0	Mass A of target nucleus (= number of nucleons). If zero, a default isotope is chosen for the given Z.
Z	integer	20	Charge Z of target nucleus (= number of protons).
Projectile_A	integer	-99	deprecated, use 'A' instead
Projectile_Z	integer	-99	deprecated, use 'Z' instead
fermiMotion	logical	.true.	Determines whether particles in target nucleus feel Fermi motion.
densitySwitch_static	integer	3	This switch is important, because it decides, which static density is used to set up the testparticles in the nuclei before the first time-step. Possible values: • 0 : density=0.0 • 1 : Static density using Woods-Saxon according to H. Lenske • 2 : Static density according to Nieves93 • 3 : Static density using Woods-Saxon according to H. Lenske, different radii for neutrons and protons • 4 : Static density according oscillator shell model • 5 : Density distribution is a sphere with density according to the input value of "fermiMomentum_input". • 6 : Static density based on LDA, implemented by Birger Steinmueller • 7 : Static density based on LDA + Welke potential • 8 : Static density prescription according Relativistic Thomas-Fermi (valid only in RMF-mode) Possible nuclei for the different prescriptions: • 1 : A > 2 (only A > 16 makes sense) • 2 : Be(9), C(12), O(16,18), Al(27), Ar(40), Ca(40,44), Fe(56), Cu(63), As(75), Ce(142), Sn(112,116,120,124), Ta(181), Au(197), Pb(208) see densityStatic.f90, subroutine denspar for more info • 3 : 6→C(12), 8→O(16),O(18), 13→Al(27), 20→Ca(40),Ca(44), 79→Au(197) 82→Pb(208) • 4: 2→He(4), 4→Be(9), 5→B(11), 6→C(12), 8→O(16) Refs: • J.Nieves, E.Oset, C.Garcia-Recio, Nucl.Phys.A 554 (1993) 509
fermiMomentum_input	real	0.225	Input value of the fermi momentum for densitySwitch_static=5 (in GeV).

anti	logical	.false.	Indicate, whether it is a anti-nucleus
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Propagation	code/propagation/propagation.f90		
delta_P	real	0.01	Delta Momentum in derivatives
delta_E	real	0.01	Delta energy in derivatives
UseCoulombDirectly	logical	.true.	Whether to use coulomb force directly in propagation or not. (If switched off while coulomb is switched on in module coulomb, the effect of the coulomb potential comes in via the gradient of the potentials. With this flag you can not switch on/off coulomb, you just select, how it is treated.)
UseHadronic	logical	.true.	Whether to use hadronic potentials in propagation
FreezeNonint	logical	.false.	If switched on, the real particles which did not interact will have zero velocities, i.e. will be "frozen". This is important for stability of the nuclear ground state in real particle simulations. Note that this flag influences only when freezeRealParticles=.false.
RungeKuttaOrder	integer	1	Order of Runge-Kutta in derivatives: • 1 = first order Runge-Kutta • 2 = second order Runge-Kuttay
Mode	integer	2	define the type of propagation: • 0: Cascade • 1: Euler • 2: PredictorCorrector • 3: no propagation, random placement in box
dh_dp0_switch	logical	.true.	Switch which decides whether we use dh_dp0.
offShellInfoDetail	logical	.false.	print out detailed offShellInfo
tachyonDebug	logical	.false.	...

propagation_RMF_input	code/propagation/propagation_RMF.f90		
predictorCorrector	logical	.true.	Switch for predictor-corrector method in the propagation. If .false. then simple Euler method is used (i.e. only predictor step is done)

deleteTachyons	logical	.false.	Switch for treatment of particles with velocity > 1 . Possible values: • if .true., these particles are deleted by setting ID=0 • if .false., these particles are propagated, but with modified velocity
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pythia	code/collisions/twoBodyReactions/HiEnergy/DoCollTools.f90	
MSEL	integer	Pythia variable
MSTU	integer, dimension(200)	Pythia array
MSTJ	integer, dimension(200)	Pythia array
MSTP	integer, dimension(200)	Pythia array
MSTI	integer, dimension(200)	Pythia array
PARU	real, dimension(200)	Pythia array
PARJ	real, dimension(200)	Pythia array
PARP	real, dimension(200)	Pythia array
PARI	real, dimension(200)	Pythia array
CKIN	real, dimension(200)	Pythia array
PMAS	real, dimension(500, 4)	Pythia array

MDCY	integer, dimen- sion(500, 3)	Pythia array
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residue_Input code/analysis/sourceAnalysis/residue.f90			
DetermineResidue	logical	.true.	If .true., then the determination of target residue properties for every event will be done. Their output in file 'TargetResidue.dat' at the end of time evolution is called elsewhere. If nothing is stored, no output is generated.
mode	integer	1	select the mode, how the residue energy is determined (field res%mom(0)): • 1: the sum of hole excitation energies • 2: the sum of energies of the removed particles (with minus sign)
switchOutput	integer	0	select the output: • 1: write out TargetResidue.dat • 2: write out TargetResidue.Plot.dat • 3: write out both files

ResonanceCrossSections code/collisions/twoBodyReactions/baryonMeson/resonanceCrossSections.f90			
fullPropagator	logical	.false.	Includes also the real parts in the resonance propagator. In former works (i.e. in the old Effenberger code) this has been neglected. It should be set to .true. only if mediumSwitch_coll=.true. in the namelist width_Baryon.

RMF_input code/rmf/RMF.f90			
RMF_flag	logical	.false.	If .true. then use relativistic mean fields.

N_set	integer	1	Select parameter set to use: • 1 — NL1 [Lalazissis] (K=211.29 MeV, $m^*/m=0.57$) • 2 — NL3 [Lalazissis] (K=271.76 MeV, $m^*/m=0.60$) • 3 — NL2 [Lang] (K=210 MeV, $m^*/m=0.83$) • 4 — NLZ2 [Bender] (K=172 MeV, $m^*/m=0.583$) • 5 — NL3* [Lalazissis, priv. comm.] (K=258.28 MeV, $m^*/m=0.594$) • 6 — Same as N_set=3, but including the rho meson. • 7 — NL1 [Lee] (K=212 MeV, $m^*/m=0.57$) • 8 — NL2 [Lee] (K=399 MeV, $m^*/m=0.67$) • 9 — Set I [Liu] (K=240 MeV, $m^*/m=0.75$) • 10 — NL1 [Lang] (K=380 MeV, $m^*/m=0.83$) • 11 — NL3 [Lang] (K=380 MeV, $m^*/m=0.70$) • 31 — Parity doublet model Set P3 [Zschesche] (K=374 MeV) • 32 — Parity doublet model Set P2 [Zschesche] (K=374 MeV) • 33 — Parity doublet model Set 1 [Shin] (K=240 MeV) • 34 — Parity doublet model Set 2 [Shin] (K=215 MeV) References: • Bender et al., PRC 60, 34304 (1999) • Lalazissis et al., PRC 55, 540 (1997), • Lang et al., NPA 541, 507 (1992) • Lee et al., PRL 57, 2916 (1986) • Liu et al., PRC 65, 045201 (2002) • Shin et al., arXiv:1805.03402 • Zschesche et al., PRC 75, 055202 (2007)
grad_flag	logical	.false.	If .true. then include space derivatives of the fields
lorentz_flag	logical	.true.	If .false. then the space components of the omega and rho fields are put to zero
Tens_flag	logical	.false.	If .true. then compute the energy-momentum tensor and four-momentum density field (not used in propagation)
flagCorThr	logical	.false.	If .true. then the srtfree of colliding particles is corrected to ensure in-medium thresholds of $BB \rightarrow BB$ and $MB \rightarrow B$
kaonpot_flag	logical	.false.	This switch turns on the Kaon potential in RMF mode
fact_pbar	real	1.	Modification factor for the antiproton coupling constants
fact_Delta	real	1.	Modification factor for the Delta(1232) coupling constants
fact_hyp	real	1.	Modification factor for the hyperon coupling constants
fact_antihyp	real	1.	Modification factor for the antihyperon coupling constants
fact_Xi	real	1.	Modification factor for the Xi and XiStar coupling constants

fact_antiXi	real	1.	Modification factor for the antiXi and antiXiStar coupling constants
fact_kaon	real	0.	Modification factor for the Kaon and antikaon coupling constants
flagVectMod	logical	.true.	This switch turns on the modification factors for vector couplings

selfenergy_realPart	code/spectralFunctions/selfenergy_baryons.f90		
rel_accuracy	real	0.05	Relative accuracy for resonance self energy
intSolver	integer	1	Decide on the numerical package to be used for the Cauchy integral: • 1=quadpack routine • 2=cernlib routine
makeTable	logical	.true.	Switch on/off the usage of an input tabulation
noDispersion	logical	.false.	Switch on/off the usage dispersion relations
maxRes	integer	100	
minRes	integer	-100	
extrapolateAbsP	logical	.false.	if(true) then set absP to maxAbsP if absP is larger
writeLocal	logical	.false.	• Tables are outputted to local directory, not to buuinput

selfEnergyMesons	code/spectralFunctions/selfenergy_mesons.f90		
dispersion	logical	.false.	Use dispersive real parts of the self energy.

SMM_input	code/analysis/sourceAnalysis.f90		
SMM_Flag	logical	.false.	if .true. then source analysis is switched on
rho_cutoff	real	100.	density cutoff (in units of the saturation density "rhoNull") which defines "emitting" particles
spectator_cutoff	real	1.	min. value of number of collisions which defines "spectator"-matter
A_cutoff	integer	2	min. value of the source mass number
SelectionMethod	integer	0	defines the selection method of spectators and fireball. Can be used in high energy Hadron-Nucleus events.
betaChoice	integer	0	Defines the way to calculate the source velocity in RMF mode. Has no influence in calculations with Skyrme potential.

MaxTimePrinting	integer	10	Indicates how many times the results are printed into files. NOTES Set MaxTimePrinting to a very big value, i.e. 1000, if you wish that the BUU-run develops until time=time_max.
DetailedHyperonOutput	logical	.true.	print more informations for Hyperons and pions.
hyperSource	logical	.false.	If true, the Lambda and Sigma0 hyperons will be included into source

spectralFunction

code/spectralFunctions/spectralFunc.f90

which_nuclwidth	integer	1	This flag decides what is used for the nucleon width. Possible values: • 1 : use constant width given in const_nuclwidth • 2 : use width increasing linear with density; $\Gamma = \text{const} \cdot \rho / \rho_0$ with const given in nuclwidth_dens • 3 : use toy model (constant NN cross section) • 4 : use realistic width (cf. diploma thesis of D. Kalok) • 5 : use realistic width: width based on our collision term NOTES The correct normalisation has not been included here!!
nuclwidth	real	0.001	if which_nuclwidth=1, nuclwidth gives the width used in the Breit-Wigner for the nucleon
nuclwidth_dens	real	0.006	if which_nuclwidth=2, nuclwidth_dens gives the width used in density dependent width 6 MeV are motivated in F. Froemel dissertation
nuclwidth_sig	real	5.5	if which_nuclwidth=3, nuclwidth_sig gives the NN cross section in fm ²
relativistic	logical	.true.	Use either relativistic or non-relativistic spectral functions.
widthMass	integer	1	select which mass is used to calculate the width: • 1: bare mass • 2: invariant mass • 3: invariant mass + mass shift of nucleon (for PDM) (= 1 for RMF and Skyrme)

spectralFunctionMesons

code/spectralFunctions/spectralFuncMesons.f90

relativistic	logical	.true.	• Use either relativistic or non relativistic spectral functions.
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target	code/density/nucleus.f90		
A	integer	0	Mass A of target nucleus (= number of nucleons). If zero, a default isotope is chosen for the given Z.
Z	integer	20	Charge Z of target nucleus (= number of protons).
Target_A	integer	-99	deprecated, use 'A' instead
Target_Z	integer	-99	deprecated, use 'Z' instead
fermiMotion	logical	.true.	Determines whether particles in target nucleus feel Fermi motion.
densitySwitch_static	integer	3	This switch is important, because it decides, which static density is used to set up the testparticles in the nuclei before the first time-step. Possible values: • 0 : density=0.0 • 1 : Static density using Woods-Saxon according to H. Lenske • 2 : Static density according to Nieves93 • 3 : Static density using Woods-Saxon according to H. Lenske, different radii for neutrons and protons • 4 : Static density according oscillator shell model • 5 : Density distribution is a sphere with density according to the input value of "fermiMomentum_input". • 6 : Static density based on LDA, implemented by Birger Steinmueller • 7 : Static density based on LDA + Welke potential • 8 : Static density prescription according Relativistic Thomas-Fermi (valid only in RMF-mode) Possible nuclei for the different prescriptions: • 1 : A > 2 (only A > 16 makes sense) • 2 : Be(9), C(12), O(16,18), Al(27), Ar(40), Ca(40,44), Fe(56), Cu(63), As(75), Ce(142), Sn(112,116,120,124), Ta(181), Au(197), Pb(208) see densityStatic.f90, subroutine denspar for more info • 3 : 6→C(12), 8→O(16),O(18), 13→Al(27), 20→Ca(40),Ca(44), 79→Au(197) 82→Pb(208) • 4: 2→He(4), 4→Be(9), 5→B(11), 6→C(12), 8→O(16) Refs: • J.Nieves, E.Oset, C.Garcia-Recio, Nucl.Phys.A 554 (1993) 509
fermiMomentum_input	real	0.225	Input value of the fermi momentum for densitySwitch_static=5 (in GeV).

ReAdjustForConstBinding	logical	.true.	If this flag is set to true, we use the selected density distribution only for a preliminary step, where we calculate the baryonic potential as function of r (which depends on the density distribution). From the condition, that the binding energy has to be constant, we deduce the distribution of the fermi momentum and thus the 'new' density distribution. The tabulated density distribution is replaced via the 'new' one and all behaviour is as usual. Only valid in Skyrme mode (and also not for HeavyIon init).
ConstBinding	real	-0.008	if 'ReAdjustForConstBinding' equals true, we a trying to readjust the fermi momentum and the density such, we quarantee this value for the binding energy (in GeV). Only valid in Skyrme mode (and also not for HeavyIon init)
radius_input	real, dimension(1:2)	-99.9	Input value(s) of the radius of the nucleus in a Woods-Saxon parametrization. If densitySwitch_static=...: • 1: one value is used both for protons and neutrons. • 3: two values: 1) proton, 2) neutron Overrides default values only if given and positive.
surface_input	real, dimension(1:2)	-99.9	Input value(s) of the surface of the nucleus in a Woods-Saxon parametrization. If densitySwitch_static=...: • 1: one value is used both for protons and neutrons. • 3: two values: 1) proton, 2) neutron Overrides default values only if given and positive.
doPrintGlauber	logical	.false.	switch on/off printing of simple Glauber calculations after init returning σ_{hA} as function of σ_{hN} (cf. Falter PhD, eq. (5.15))

TransportGivenParticle	code/init/initTransportGivenParticle.f90		
particle_ID	integer	1	Determines what kind of particle is initialized (see idTable)
charge	integer	1	Determines what charge
position	real, dimension(1:3)	(/0.,0.,0./)	Determines the position.
threemomentum	real, dimension(1:3)	(/0.,0.,1./)	Determines the three-momentum.
mass	real	-1.	Determines the mass (if negative, choose mass according to spectral function).
maxmass	real	1.5	Determines the maximum mass (if mass is chosen according to spectral function).

perweight	real	1.	Determines the weight.
frequency	integer	10	after this amount of time steps a new output file is generated
initRandomRadiativeDelta	logical	.false.	intended use: radiativeDelta decay. chooses position,threemomentum,mass of Delta randomly; charge is choosen either 0 or 1

W_distributions code/analysis/neutrinoAnalysis.f90			
dW_Npi	real	0.02	for dsigma/d(InvariantMass); only work if include_W_dist is .true. set the min, max and steps for various W-distributions
Wmin_Npi	real	1.08	for dsigma/d(InvariantMass); only work if include_W_dist is .true. set the min, max and steps for various W-distributions
Wmax_Npi	real	1.6	for dsigma/d(InvariantMass); only work if include_W_dist is .true. set the min, max and steps for various W-distributions
dW_mupi	real	0.04	only work if include_W_dist is .true. set the min, max and steps for various W-distributions
Wmin_mupi	real	0.24	only work if include_W_dist is .true. set the min, max and steps for various W-distributions
Wmax_mupi	real	1.2	only work if include_W_dist is .true. set the min, max and steps for various W-distributions
dW_muN	real	0.04	only work if include_W_dist is .true. set the min, max and steps for various W-distributions
Wmin_muN	real	1.04	only work if include_W_dist is .true. set the min, max and steps for various W-distributions
Wmax_muN	real	2.12	only work if include_W_dist is .true. set the min, max and steps for various W-distributions

width_Baryon code/width/baryonWidthMedium.f90			
mediumSwitch	logical	.false.	Switch on and off the in-medium width of all baryons at once. If .false., the vacuum width are used.
mediumSwitch_Delta	logical	.false.	Only meaningful if mediumSwitch=.true.: Switch on and off the in-medium width of the Delta. (.false.=off) Note that in that case the Delta is treated specially: what is used for the in-medium width is determined by the flag in deltaWidth. This switch is not consistent with mediumSwitch_coll!

mediumSwitch_proton_neutron	logical	.false.	Only meaningful if mediumSwitch=.true.: Switch on and off the in-medium width of the proton and the neutron. (.false.=off) Note that in that case the nucleons are treated specially. This switch is not consistent with mediumSwitch_coll!
mediumSwitch_coll	logical	.false.	Only meaningful if mediumSwitch=.true.: Use in-medium width according to collision term. NOTES if set to TRUE, then also UseOffShellPotentialBaryons (see module offShellPotential) must be .true.
verboseInit	logical	.false.	switch on/off informational messages during initialization
verboseInitStop	logical	.false.	Stop after informational messages during initialization or not.

width_Meson		code/width/mesonWidthMedium.f90	
mediumSwitch	integer	0	Treatment of In-Medium Widths for mesons: • 0: Only vacuum widths are used. • 1: The collisional width is assumed to be constant (only density-dependent). • 2: The full tabulated in-medium width is used, as calculated via the collision term. Isospin asymmetry of nuclear matter included. Zero temperature assumed. All mesons are in-medium broadened. • 3: Same as 2 but for isospin symmetric nuclear matter at finite temperature. Only rho-meson is in-medium broadened. Other mesons not modified.
Gamma_coll_rho	real	0.150	Collisional width for the rho meson in GeV. Only used if mediumSwitch = 1.
Gamma_coll_omega	real	0.150	Collisional width for the omega meson in GeV. Only used if mediumSwitch = 1.
Gamma_coll_phi	real	0.030	Collisional width for the phi meson in GeV. Only used if mediumSwitch = 1.
verboseInit	logical	.false.	switch on/off informational messages during initialization
allowMix	logical	.false.	switch on/off linear interpolation between bins in density while returning the tabulated values for MassAssInfo.

XsectionRatios_input				code/collisions/phaseSpace/XsectionRatios.f90
flagScreen	logical	.false.	• If .true. – in-medium screening is applied to the input cross section. • If .false. – no cross section modification. NOTES flagScreen should be replaced by flagInMedium	
ScreenMode	integer	1	possible values: • 1: in-medium screening of NN total cross section according to Li and Machleidt, PRC 48 (1993) 1702 and PRC 49 (1994) 566 • 2: in-medium screening of BB total cross section according to P. Danielewicz, NPA 673, 375 (2000); Acta. Phys. Pol. B 33, 45 (2002) NOTES relevant when flagScreen = .true., is deprecated	
flagInMedium	logical	.false.	• If .true. – In-medium ratios are used to decide whether an event is accepted or not. • If .false. – The event is always accepted	
InMediumMode	integer	2	possible values: • 1: all events of the type $BB \rightarrow BB$ (+ mesons) are subject to in-medium reduction following Eqs.(194),(195) of GiBUU review paper [currently works in RMF mode only] • 2: $NN \rightarrow NN$ elastic scattering events are modified according to Li and Machleidt all other $BB \rightarrow BB$ (+ mesons events are subject to in-medium reduction according to Eq. (33) from T. Song, C.M. Ko, PRC 91, 014901 (2015) [works in all modes (Skyrme, RMF, cascade)] Elastic cross section change only can be obtained by setting $\alpha = 0$ • 3: Only Song-Ko modification for Delta excitation turned on NOTES relevant when flagInMedium = .true.	
flagVacEL	logical	.false.	• If .true. – no in-medium modification for $NN \rightarrow NN$ elastic NOTES relevant for flagInMedium = .true.	
flagVacHRES	logical	.false.	• If .true. – no in-medium modification for $BB \leftrightarrow BB$ with at least one participating resonance higher than P33(1232) or with more than one P33(1232) NOTES relevant for flagInMedium = .true.	
flagVacMesProd	logical	.false.	• If .true. – no in-medium modification for $BB \rightarrow BB + \text{meson(s)}$ NOTES relevant for flagInMedium = .true.	

alpha	real	1.2	Parameter which controls the density dependence of the NN $\leftrightarrow$ N Delta cross section via suppression factor of $\exp(-\alpha \cdot (\rho/\rho_0)^{\beta})$ for the density dependence from: Song/Ko, arXiv:1403.7363 (InMediumMode=2 or 3)
beta	real	1.	Parameter which controls the density dependence of the NN $\leftrightarrow$ N Delta cross section via suppression factor of $\exp(-\alpha \cdot (\rho/\rho_0)^{\beta})$ for the density dependence of the type of arXiv:2107.13384 (InMediumMode=2 or 3)
shift0	real	0.	Mass shift $m-m^*$ (GeV) for using in elementary particle collision mode.

YScalingAnalysis

code/analysis/yScalingAnalysis.f90

analyze	logical	.false.	Determines wether the y-scaling analysis is performed
optionalOutput	logical	.false.	Determines wether in addition to the standard 'scaling_analysis.dat' other histograms will be generated. E.g. * 'single_nucleon.dat' - a table for comparing nucleon-knockout with fully inclusive <pre> jpre, x sections * 'scaling_info.dat' - general parameters of the analysis, to be used for quick jpre, analysis * 'scaling_delta.dat' - output to be used for analysis of scaling function in jpre, resonance excitation region </pre>
variable	integer	1	determines which kind of scaling variable will be used (cf. Donnelly, Sick 1999): • 1) RFG full variable Psi • 2) RFG approximation Psi • 3) PWIA full Upsilon (y/k_f) • 9) evaluation will be done for all variables, output written to seperate files
kFermi	real	0.2251	Nucleon Fermi momentum in nucleus. If none specified 0.2251 will be used, except if densitySwitch_static is set to 5, then fermiMomentum_input is used. The 0.2251_ aims at preventing confusion whith delibaretely set differences between kFermi and fermiMomentum_input
E_shift	real	0.020	Energy correction to account for binding effects, otherwise neglected in RFG model

Yukawa code/potential/yukawa.f90			
yukawaFlag	logical	.false.	Switches Yukawa potential on/off
smu	real	2.175	Yukawa mass in fm**(-1). (range of potential)